

Education Applications

Take a look inside almost any classroom in an elementary school, high school, university, or a corporate training room and what will you find? Most likely, the communication of visual phenomenon is attempted using some combination of two-dimensional display devices such as textbooks, chalkboards, overhead projectors, bulletin boards, and computer screens.

Although many scholastic subjects involve learning about three-dimensional entities, the pupil's primary mode of receiving information is two dimensional. With the exception of certain cases where physical artifacts (for example arrowheads, or ball-and-stick models of molecules) are integrated into the instructional program, the student must extend the ideas that are being portrayed on their textbook pages or computer screens into three dimensions entirely within their own minds. However, research performed by Chris Dede and R. Bowen Loftin indicates that this is not an effective way to learn truly three-dimensional concepts. Often students walk away from the classroom only able to discuss the phenomena as though it were only two-dimensional in nature, unable to expand concepts into 3D.

In addition to the constraints of presentations limited to two-dimensions, common classroom displays are also not interactive. Some (such as movies) incorporate motion, and some (computers) can incorporate interaction. Virtual reality, however, provides a true three-dimensional, interactive environment in which students can manipulate, explore, and modify their environment.

Additionally, virtual reality offers the ability for students to collaborate with others who may be located elsewhere, and VR can provide anonymity for the students when that is desirable. Finally, virtual reality offers the ability to record student performance and actions, allowing either the computer or teacher to respond appropriately.

The related area of augmented reality provides many capabilities that are useful in educational settings. Augmented reality is the medium in which synthetic

imagery and information are overlaid on top of the real world. An example application would be to allow a student to take a virtual nature walk through a grove of trees. In this difficult-to-implement example, as they walk and look around, students would be able to see annotations describing the tree they are seeing, as well as offered interactive quizzes, animations, and other practical educational opportunities.

One reason that virtual reality has not been widely adopted in schools is that there are relatively few appropriate applications available; another is the cost of the required equipment. These two issues are clearly related. Why develop applications if schools cannot afford to implement them in their curriculum?

Fortunately, these two problems—hardware cost and software availability—are becoming less of a barrier. Low-cost, commodity VR hardware has brought system cost down an order of magnitude over systems such as those prominent in this book. Open-source hardware platforms such as the *GeoWall* allow users to construct their own systems.

Companies such as Visbox, Inc., VisTek, Inc., and CAE-net (John-E-Box) provide relatively inexpensive commercial turnkey systems. Certainly as large-screen high-definition displays and computer vision-based camera inputs become increasingly pervasive, such systems will involve only modest costs above that of a typical classroom computer—indeed, such technology is already becoming standard in home gaming systems. Software for education is being addressed in a number of open-source efforts.



FIGURE 6-1

Third-grade Cub Scouts enjoy a tour of the VisBox-X2 system at Valparaiso University.

Visualization software for chemistry includes *Ribbons* (University of Alabama, Birmingham) and *VMD* (University of Illinois). A host of software applications covering education in engineering and mathematics is available from the Scientific Visualization Laboratory (SVL) at Valparaiso University. One of the SVL's applications (called *MaxwellVU*) was directly inspired by the *ScienceSpace* application detailed below. This application is open source, available on a number of low-cost hardware platforms, and allows students to experience and interact with electromagnetic fields in a virtual environment. The goal is to communicate the inherent meaning of Maxwell's equations relating electric and magnetic fields.

6.1 AREAS OF APPLICATION

Virtual reality is currently being used or tested in a variety of formal and informal educational settings. Formal settings include elementary school, secondary school, universities, etc. Informal settings include museums, zoos, libraries, and other similar venues. We have categorized these applications into four areas, and describe at length at least one example for each:

- **Sciences** In this chapter we cover the application *ScienceSpace*, which allows students to explore difficult three-dimensional science problems ranging from electromagnetics, to kinematics, to chemistry.
- **Constructivist** Virtual reality is particularly conducive to manifesting constructivist theories of education in that it allows students to learn as they build their own environments. This chapter describes several constructivist applications that cover a variety of student age groups.
- **Informal Education** Around the world, students are using virtual reality to learn about topics ranging from gorilla social behavior to ancient civilizations. This chapter describes several informal education settings.
- **Special Education** At a very practical level, virtual reality has been used to train critical life skills to students with special needs. Virtual reality offers a safe place to practice new skills and behaviors, and simultaneously allows instructors to gather data to help assess the progress of their students and provide new training in response. This chapter includes descriptions of virtual reality applications to help develop training skills such as manipulating a powered wheelchair, using a city bus system, and basic life-skill and social-skill training for students with autism.

6.2 OTHER AND FUTURE USES

A trend that is already emerging in the area of education is the use of augmented reality.

We envision more adoption of AR technologies in areas ranging from providing a more personalized experience for museum visitors to providing students guided exercises in prairie and other natural environments. AR offers the potential to provide extra information and annotations directly about the phenomena the student is experiencing at that moment in time, and tailored to their own interests and learning level.

VR and AR technologies are beginning to blur the line of where one “goes” to go to school. VR, AR, and telepresence allow students to experience many environments, and collaborate with geographically disperse students, scientists, doctors, and other experts while at their school, or potentially even while at home.

The possibilities of simulations carried out in virtual environments allow for compelling historical re-creations (allowing students to participate in historical events), science expeditions, and any learning experience that can make use of simulations and vignettes. Such applications can teach students emergency preparedness, social skills, and practical skills such as learning to use a new scientific or musical instrument.

CASE STUDY 6.1: PROJECT SCIENCESPACE

Executive Summary

Traditionally, scientific phenomena are presented to students as two-dimensional visualizations through a few different mechanisms: photos and diagrams in books, drawings on blackboards and overhead projectors, and more recently, interactive simulations on computer screens. All these displays have something in common. They are all two dimensional. Two-dimensional imagery has been shown to be a reasonable teaching tool, even for three-dimensional concepts, but further studies have shown that, while students may be able to answer test questions correctly, they often don't understand how those concepts extend into three dimensions as they do in nature.

Researchers at George Mason University, in collaboration with the University of Houston, developed an experimental test bed in which they studied the use of three-dimensional virtual reality for teaching three-dimensional concepts. As an experimental platform they developed three worlds for teaching concepts of physics and chemistry. *NewtonWorld* helps students explore concepts of Newtonian mechanics. *MaxwellWorld* provides an interactive space for studying concepts of electromagnetic fields, and *PaulingWorld* is focused on interactions between molecules. Collectively, the research is called *Project ScienceSpace*.

Although other applications of virtual reality in education have been explored, *ScienceSpace* specifically focused on studying the effectiveness of using virtual reality for teaching science concepts. The target audience is grade school, high school, and undergraduate students. Because most schools don't have large, projection virtual reality facilities, the researchers have developed a system that uses portable head mounted displays. Besides looking at the effectiveness of VR in general for teaching science, the researchers also explored the possible benefits of displaying to multiple senses (sight, sound, and touch).

Results of the studies indicate that immersive environments are more effective for teaching three-dimensional science concepts, and that displaying to multiple senses is more effective than displaying to a single sense.

Introduction

Trends in technology suggest that VR will soon be in the hands of a growing population of computer users. Heretofore,

access to virtual reality facilities has generally been limited to large research centers and individual efforts funded to test the usefulness of the technology on certain populations. To examine and prepare for a time when VR will be widely available in grade school, high school,¹ and college classrooms, collaborators at George Mason University and the University of Houston produced a series of virtual reality experiences designed to test VR as a potential educational resource.

The focus of their work is on science education. Science was chosen because of an established teaching methodology and a well-known lack of overall student comprehension. *Project ScienceSpace* consists of three separate experiences designed to teach through scientific inquiry: *NewtonWorld*, *MaxwellWorld*, and *PaulingWorld*.

The goal of the *ScienceSpace* effort was to lead to better educational tools that will help students learn by allowing them to experience concepts directly, rather than merely memorize formulas and abstract behaviors. By directly interacting with the laws of physics, it is hoped that they will accept them as a valid explanation of natural phenomena, rather than as some arcane formulation presented by their teachers. It has been observed that students who ably solve physics problems on a test behave in day-to-day activities as though these concepts are not true.

The specific objectives of these studies were threefold:

- To determine whether physically immersive, constructivist learning can remedy typical misconceptions of reality held by many students
- To enhance mastery of abstract scientific concepts (e.g., relativity, quantum mechanics, molecular-orbital bonding) in an experiential, physically immersive, multisensory, collaborative environment
- To investigate the degree to which the effectiveness of using virtual reality for education is undermined as a result of physical or psychological effects caused by VR (e.g., disorientation)

¹For non-U.S. readers: Elementary school children generally range from 5 to 14, and high school students from 14 to 18 years old.

The traditional method of teaching science through lectures, books (with twodimensional diagrams), and laboratory experiments (where the participation level of many of the students is limited) works only to a degree. Yet, even students who score well often don't fully comprehend the concepts they've "learned." Furthermore, this quantitative method of teaching requires that higher mathematical skills be learned before the student can pursue the scientific concepts. The result of this circumstance is that advanced science is not being taught until after many students have already lost interest in the topic.

Initial studies targeted high school and college students with a demonstrated mastery of the concepts. Their analysis of these experiments indicates that students do learn some concepts better in VR, and can spontaneously describe concepts not yet formally introduced to them.

Application Description

The *ScienceSpace* researchers did not aim to merely develop one or two pedagogically useful applications, but rather to "map the territory" of how VR can be an effective medium for education. Their hypothesis was that VR will be useful for science education because it addresses four vital components of the nature of learning:

- The novelty of VR draws the students' attention.
- The computer worlds can be used to generate meaningful representations.
- Information can simultaneously be presented in a variety of forms.
- Students can learn by doing, using "reflective inquiry."²

The three experimental worlds all have their own unique characteristics designed to highlight particular scientific concepts. However, they also have features in common. For example, a two-handed menu-in-hand interface is used to make selecting options more natural. The first two worlds of *ScienceSpace* (*NewtonWorld* and *MaxwellWorld*) were used in the formal analysis.

NewtonWorld is designed to teach Newtonian mechanics to beginning physics students. In *NewtonWorld*, Newton's laws of kinematics and dynamics are presented along one dimension, allowing the user to move freely outside this

dimension to view the concepts while they themselves are unhindered by Newton's laws. The concepts are presented via one or two balls that move down a colonnaded corridor with walls at each end.

Balls can be launched, caught, and "ridden" by students, who can control parameters of the world such as the mass of the ball and the magnitude of forces applied to the ball. *NewtonWorld* balls exhibit conservation of linear momentum, and demonstrate concepts of kinetic energy, as well as Newton's Laws of Motion.

By providing multisensory cues of important components of the world, students have a better opportunity to focus on the relationships between them. Examples of multisensory representations include potential energy (represented both haptically and visually) and velocity (represented aurally and visually).

MaxwellWorld is designed to allow students to explore the nature of electrostatic forces and fields, and to understand the concept of electric flux. The content of *MaxwellWorld* is a collection of point electrical charges represented as spheres, with the resulting field lines and equipotential surfaces represented as lines and surfaces.

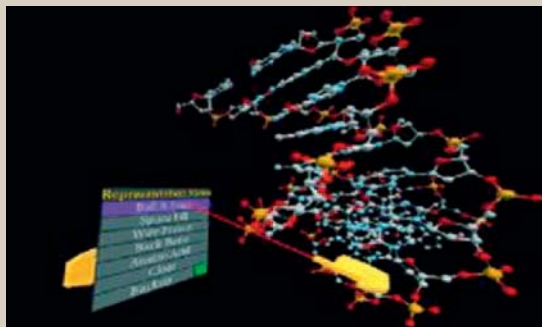
Holding a tracked wand, students create and place electric charges in space, producing representations of the electric forces, fields and flux. Exploration of the electromagnetic properties can be done by visual inspection, in conjunction with other techniques such as probing calculated numeric values within the fields, etc., and even riding a test charge as it moves through the field.

PaulingWorld was created for the exploration of molecular structure and dynamics. Students can control their view of the molecular structure to examine it from any vantage point. They can also control how the molecule is represented, using the menu system to switch between various standard chemistry representational techniques. When multiple molecules are displayed, the student can witness the dynamics between them.

Representation of the Virtual World

The virtual worlds of *ScienceSpace* provide both common representations of introductory science concepts. By basing the representations on traditional two-dimensional representations, students can more easily relate their VR experience to the traditional representations they encounter.

²Dede et al.

**FIGURE 6-2**

In *PaulingWorld*, a similar interface is used to explore molecular interactions.

Each of the *ScienceSpace* worlds occupies a small region of space where the concepts are presented and the participant can maneuver. While the worlds are small enough that the participant can choose a vantage point that allows them to see the entire space from an exocentric point of view, they can also move into the world and view it egocentrically.

Abstract representations are used for all the concepts shown in the *ScienceSpace* worlds. This is necessary since the majority of the concepts cannot be directly witnessed in day-to-day life. While we can witness the indirect results of molecular docking or electric charge between two points, we cannot witness the docking itself, or the distribution of forces and energy in the charged field space. The exceptions to this are collisions in the real world, but to allow participants to better focus on the concepts, these collisions have been abstracted to occur in a one-dimensional space with no friction, and adjustable parameters.

In addition to the representations of the science concepts, menus and avatars of the user's hands are rendered as part of the user-interface, allowing the user to more easily interact with the worlds.

In *NewtonWorld*, participants view elastic collisions between two balls and between those balls and the end walls of the corridor they occupy. The corridor provides visual and aural cues of position and velocity. Columns along the corridor flash and emit a sound as a ball passes it, allowing the student to detect visual and aural patterns

**FIGURE 6-3**

NewtonWorld allows students to observe and participate in elastic collisions with virtual balls.

related to the ball's trajectory. Shadows of the balls are rendered directly below them to help the viewer gauge their movement as they pass between lines that connect the columns on either side of the corridor. A sign on each wall indicates the presence (or lack thereof) of gravity and friction.

In the initial version of *NewtonWorld*, the size of the shadows represented the amount of kinetic energy of the ball, while potential energy (of a ball being held by the user) is represented visually by a coiled spring, and haptically as vibrations in the chest of the participant. The color of the balls represented their mass, but the researchers found that this representation didn't work well, so color was removed.

After the initial experiments using *NewtonWorld*, it was redesigned with more intuitive representations of mass, and a visual cue for elasticity of each ball was added.

In *MaxwellWorld*, a 1-meter-per-side cube with Cartesian axes provides the boundary within which Maxwell's principles are represented. Source and test charges are represented as spheres with color indicating the polarity, while pulse rate of the sphere indicates the magnitude of the charge. The force on a positive test charge is represented either by an arrow emanating from the charge or by the charge's movement through the field. In "probe-mode," the user's avatar's fingertip represents the current "potential" as

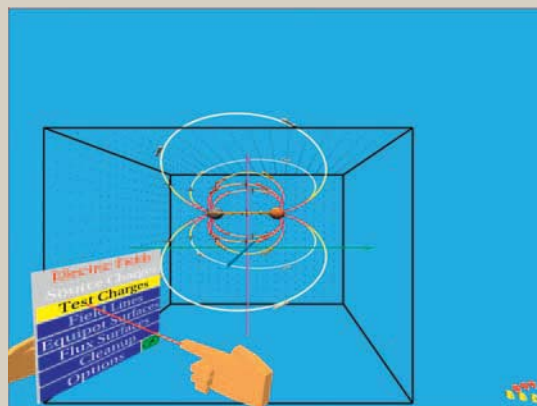
**FIGURE 6-4**

In this image of the updated *NewtonWorld*, the amount of kinetic energy contained in a ball the participant is holding is visually represented by the amount of compression of a spring attached to the ball.

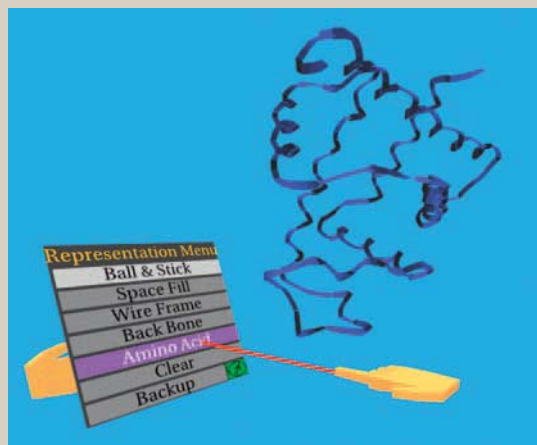
a colored plus or minus sign. The color indicates the magnitude of the potential. Similarly, a small, positive test charge is attached to the user's virtual hand with an arrow emanating from the charge indicating the direction and magnitude (again by color) of the "force [acting on] the test charge (and, hence, the electric field)." Magnitude of force was represented by the shades of color from white (neutral) to red (high), with yellow and orange in between. Potential energy was represented using the colors ranging from white to blue to green to purple.

Electric fields are represented as lines that trace a path between positive and negative source charges. To investigate Gauss' Law, the user can create spherical "Gaussian surfaces" and directly measure the flux by counting the flux lines through the sphere. Electric potentials can be represented by an equipotential surface. Field lines and equipotential surfaces are colored based on the magnitude of the force along the line or surface using the same white–red scale.

In *PaulingWorld*, molecules (represented in a variety of forms) float before the user. These forms include the familiar ball-and-stick model, a protein-backbone representation, van der Waals spheres, amino acid icons, and a wire-frame representation. Representations can also be combined, such

**FIGURE 6-5**

In *MaxwellWorld*, the participant can affect the virtual world through interactions with a menu and by placing objects and probing within the simulation space.

**FIGURE 6-6**

Several traditional molecular representations are available in *PaulingWorld*; they can be changed via a menu interface.

as using a backbone ribbon of the primary protein chain, and adding ball-and-stick representations of atoms that emanate from the backbone. The user can interactively adjust the representation, allowing them to expand a particular molecule into its ball and stick representation, for instance.

In addition to the *NewtonWorld* aural cues that co-represent the movement of the balls, sound is used in *NewtonWorld* to display the time of collisions between the balls, and the balls and the walls. In *MaxwellWorld*, sounds were used to represent the intensity of force or energy at a particular point. Sounds were also spatialized, so the user could localize in 3D the locale where the generating entity resides.

In all three worlds aural cues are used to provide feedback to the interaction interface. A sound is made whenever an item in the menu is selected, or when an object in the world is selected (e.g., to move a source charge).

The *ScienceSpace* development team also investigated the use of a variety of haptic feedback devices for use with one or more of the worlds. Some experiments with haptic devices included the use of a finger vibrator attached to a glove input device, an air-bladder glove to produce pressures on the surfaces of the hands, an exoskeleton glove to constrain hand motions when contacting a virtual object, and a glove that produces temperature changes to the fingertips. They chose not to use a mechanically linked force feedback device (such as the PHANToM) because of the restrictions it places on the usable working volume where interactions can take place.

In *NewtonWorld* tests were conducted to compare the possible benefits of multimodal representation of phenomenon. The comparison was between concepts represented visually; visually and aurally; and visually, aurally, and haptically. In this case the haptic representation consisted of a vest worn by the subjects that displays a “thump” on their chest or back when the ball ridden by the subject collided with the other ball or a wall. The results of the test indicated that with more forms of sensory feedback provided, the subject’s performance improved. However, they did not compare the effect of each individual sensory stimulation against the others.

Use of other senses such as vestibular and olfaction (smell) were not considered for inclusion with *Project ScienceSpace* because of the added complexity and expense involved.

Interaction with the Virtual World

The *ScienceSpace* project clearly falls within the genre of *science education*. Specifically, education acquired through experiential concept learning. The form of interaction is

primarily done via selecting options from a menu-in-hand interface with several methods-of-travel used, as well as by indicating desires by using a tracked single-button wand, and direct interaction using a tracked Polhemus one-button stylus.

The intended way in which the *ScienceSpace* applications would be used was to present the student with a lesson plan that included a series of exercises on the concepts, and then give them access to the immersive virtual world as the means of exploring each given problem.

Because the user’s head was fully tracked, one means of traveling through the world that was always available was by simple body movement (physical locomotion). However, each world also provided other means of travel control that were more commonly used to navigate through the worlds. All worlds provided a menu option that allowed the user to fly in the direction they pointed a handheld prop (a Polhemus stylus) while pressing its button. In this point-to-fly mode, travel speed was constant and not controllable by the user. Because point-to-fly travel was a frequent activity, all layers of the menu interface included the travel option. The travel selection was indicated by selecting a plane icon placed in the lower-right corner of each menu.

In *NewtonWorld*, if a student points at a camera icon, they are instantly teleported to that location. In other words, these icons serve as predefined interesting observation points. In *MaxwellWorld* and *PaulingWorld*, the grab-the-world method of travel was also available by selecting that option from the menu. In this mode, users can reorient or relocate the world with respect to themselves. *MaxwellWorld* also allows users to change their scale to allow them to view the world either egocentrically or exocentrically. Additionally, both *NewtonWorld* and *MaxwellWorld* offer the ability to ride-along through the world on a ball or test charge so the student can experience the forces acting on the world first hand. So, in *NewtonWorld*, you can “be the ball.”

Most of the interactions with the *ScienceSpace* worlds come from manipulating various objects in the world or general characteristics of the worlds. A menu system was the primary method of making modifications. In *NewtonWorld* (the first world built), a menu was placed as a head-up display, and users could interact with the menu by pointing at it and pressing a button to “pull-down” a list

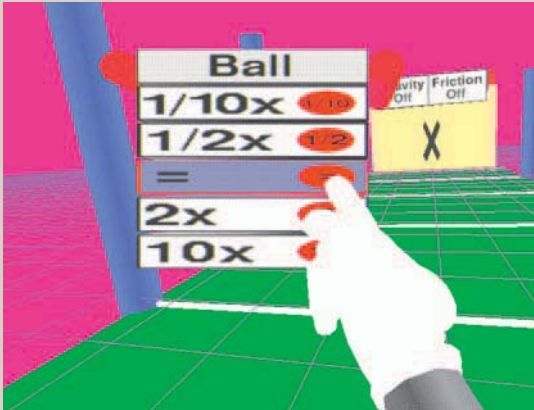


FIGURE 6-7

Two-handed menus were implemented in NewtonWorld after discovering that pull-down menus occluded important objects in the world.

of options. Having the menu constantly occupy part of the user's view was found to be intrusive and interfered with immersion, so for later worlds, the designers chose to use a bimanual (two-handed) interface for the menu.

DESIGN CHOICE

Interaction

The fixed-position menu used in *NewtonWorld* was abandoned for a bimanual menu interface for later worlds because the *NewtonWorld* users found the menu to be distracting and intrusive. By attaching the menu to the off hand, users could then eliminate the menu from view simply by putting their hand by their side, yet they could quickly and easily access the menu by raising their hand into their field of view. Because users are able to proprioceptically sense where their hand is, they always know where the menu is, without the need for keeping it in sight.

The menu system was controlled by using physical inputs from handheld props. The user uses one prop to hold the menu in their off hand, while the second prop allows

pointing, and activation via a button press. During development of the system, the creators also experimented with other forms of input such as gesture and voice control. While they found that users liked voice control, they felt that it would add too much complexity to the already large amount of new technology they would be introducing to the classroom, so they chose not to pursue it further for this project. Also, gesture recognition was found to be unreliable, and distracting to the users, so the physical prop inputs were kept as the means of control.

In *MaxwellWorld*, users can place positive and negative charges in the world. Once these charges are placed, the virtual electric field is generated, allowing students to further interact with the resultant field. The menu system is used to choose between “probe mode,” “trace mode,” etc.

A probe showing the potential at any location can be attached to the user's virtual hand. Also, tracers can be released into the electric field causing the released tracer object to flow along as governed by the existing electric field. Electric field lines can also be individually created and manipulated. Signaling the creation of a new field line causes a line to be generated from the current probe location. Grabbing an existing line causes it to change reflecting the potential passing through the new point. When the participant activates the generation of a Gaussian surface, the surface is generated from the current probe point. Likewise, in *PaulingWorld* representations of the molecules (and atomic surfaces) are added and adjusted using the menu interface.

The *ScienceSpace* worlds described here are not populated by agents that communicate with the participant. However, the developers later investigated the possibility of integrating some of their previous work on intelligent tutoring systems to allow students to participate in the experience without requiring a human guide. Future editions could use such agents as a form of “intelligent coaches” for the learners.

The developers have also implemented a collaborative version of *MaxwellWorld* and tested it in sessions between the University of Houston and the University of California at San Diego. However, this version was not evaluated for pedagogical benefits.

The physics of these worlds were obviously created to mimic certain subsets of formulae that describe a particular

scientific theory (e.g., Newtonian physics in *NewtonWorld*). However, these physics did not necessarily apply to the user, or the form of travel they used, unless they were riding some object that was the physics model. So, features such as gravity and collision detection were only applied to specific objects to help demonstrate a particular branch of physics (e.g., Newtonian).

Venue

Thus far, the execution of the *ScienceSpace* worlds has been limited to university research labs. However, this severely limits the researchers' ability to readily engage the participation of the target subject groups (elementary school and high school students). Fortunately, as the cost of computing and VR hardware declines, constraints such as this for future educational programs will surely fall away, except in the case of *CAVE* displays, which will likely continue to be cost prohibitive for primary and secondary educational institutions. Five systems were prepared for classroom deployment to allow the researchers to get beyond the limitations of laboratory settings.

In fact, among the collaborators in this project, the University of Houston had both HMD and *CAVE* displays within its facility, and could run the worlds in either. However, at George Mason University, where many of the experiments were held, only head-mounted displays were available.

Participants of the system typically would be given some time to independently learn the science being presented, followed by time in the VR experience to either freely explore the world, or perform some specified task. Time spent in the virtual world was typically about 30–45 minutes followed by a 15-minute break, then another 30- to 45-minute session.

VR System

Since the primary venue for *ScienceSpace* is the classroom, the selected VR system must accommodate a classroom situation. However, it must be portable enough to be moved from school to school (at least until more schools have their own virtual reality facilities). Thus, an HMD-based system was constructed. In the prototyping stage of the application Virtual Research VR-3™ model HMDs were used, but these were replaced with the VR-4 model for the evaluation experiments.

The system was designed to be used while sitting down. Users can move physically by rolling in the chair, but the applications were designed such that this isn't necessary. The travel controls are typically used to navigate through the virtual worlds.

Although the HMDs match the portability and cost constraints of the project, there are some negative factors associated with them. First, the isolation factor of HMDs makes lesson administration more difficult. Students can't see written instructions or write down notes. They can converse with an instructor, but they can't communicate visually. Also, HMDs are not generally very comfortable, and less costly models (as those used in this project) are still low-resolution devices making for a less than auspicious experience.

At the University of Houston, where a *CAVE*-like projection display was installed, this application was also ported to the *CAVE*. Anecdotal experience indicates that experiencing *MaxwellWorld* in the *CAVE* is better for seeing the world from the exocentric perspective, while egocentric viewing is easier in the HMD.

During the initial development and evaluation, the computation and graphics system was a Silicon Graphics Onyx with Reality Engine2 graphics hardware. A Silicon Graphics Indy™ desktop workstation provided the audio output. This system (the Onyx in particular) was not very amenable to transporting from location to location. By using higher-performance machines during the experimental analysis stage of the project, the developers did not have to concern themselves with tweaking the software to squeeze every ounce of performance out of the hardware. They had the luxury of knowing that by the time such a system would be deployed in schools, the hardware would have improved sufficiently that they would be safe with reasonably well-written programs, and thus could spend their time developing the interface and representations aspects of the software. The systems assembled for classrooms used a single Silicon Graphics O2 system for computation, graphics, and audio rendering. The O2 system is quite small, and readily portable. Today, even a basic laptop has sufficient computational and graphical power to handle these experiences. Position tracking was done with a Polhemus electromagnetic tracking system (the *CAVE*-like projection display used an Ascension Flock of Birds electromagnetic system). Three

physical items were tracked: the user's head, a 3-Ball or stylus tracked input device (with a button switch), and a third device the developers refer to as a "fixture." The "fixture" device is basically used to provide a means for allowing the user to position the menu in the virtual world.

Although not part of the "main" system, several other display devices were experimented with to render various forms of haptic sensations. Haptic displays for presenting vibrations or thumps to the chest, vibrations to the hand, or temperature changes to the hand were all forms with which they experimented.

Vibrations, or "thumps," could be delivered to the chest by vests equipped with special hardware. Two types of vests were used: a commercial off-the-shelf (COTS) device designed for use with TV video games was interfaced to the system, and a "homemade" model was created using a hunting vest and woofer speakers.

Other specialized hand-displays were used for haptic display. These glove-displays included a vibrator glove from Virtual Technologies, the Teletact bladder glove, a Rutgers Dextrose Hand-Master, and a Displaced Temperature Sensing System (DTSS) display glove from CM Research. A PHANToM mechanical-arm-based force display was also considered, but concerns about the small working volume of the PHANToM led to the developers choosing not to explore it for this application.

For experimenting with the use of voice for interaction control, the developers used the "Wizard of Oz" approach to input, where someone sat at a keyboard and entered commands based on what the user said.

Application Implementation

A collaboration between the University of Houston (UH) and George Mason University (GMU), *Project ScienceSpace* began in February 1994. Initial experiments were run in *NewtonWorld* in that summer, with experiments using *MaxwellWorld* beginning in 1995. The development of *PaulingWorld* extended later.

Development was shared between UH and GMU, with the technical development conducted at the Virtual Environment Technology Laboratory (VETL) at the University of Houston, and the pedagogical and human factors design concepts developed at GMU in Fairfax, Virginia.

Development of the project at two sites significantly distant from one another was initially done using teleconferenced conversations and faxed diagrams. Once the prototype system was running in Houston, a duplicate system was developed in Fairfax, allowing either site to run the application. The ability of both development sites to run the experience simultaneously allowed collaborative design/debugging meetings to be conducted via normal telephone communications.

The software was developed using the VR-tool software system previously developed at the VETL in Houston. The VR-tool system was built upon Silicon Graphic's *Inventor* toolkit, and used Silicon Graphics workstations. Writing the applications with *Open Inventor* enabled them to be easily portable to other platforms.

The primary design concerns addressed the use of *ScienceSpace* worlds as learning experiences. The primary focal points of design then were on user interface issues, such that the applications would be as useful as possible.

The design process involved several iterative cycles of design and evaluation. Evaluation was done by preliminary tests of small groups of students as well as by allowing a large group of physics educators to experience and comment on the project during an educational forum (the 1994 Summer Meeting of the American Association of Physics Teachers). After the initial worlds were designed and experiments conducted, the developers chose to address a younger audience, and used focus groups to evaluate and brainstorm new ideas for the experiences. They also used "talk-aloud protocols" to evaluate how the user experienced the worlds by listening to their thought process as they performed various actions.

Design of later worlds (*MaxwellWorld* and *PaulingWorld*) improved upon the design of *NewtonWorld*. For example, the menu system that had been fixed in the HMD was now dynamically positioned with the user's off-hand. Also, *NewtonWorld* was later revamped to make use of lessons learned by the experiments with both *NewtonWorld* and *MaxwellWorld*. In addition to improving the menu system, many representational idioms were modified from the initial *NewtonWorld* system to better present the concepts. For example, instead of representing mass as varying (arbitrary) colors, the level of transparency is varied from high

transparency (lower mass) to near opacity (high mass). Elasticity was indicated by the surface texture changes from shiny (representing hard, elastic balls) to a “soft and squishy” (bubblegum) look.

VR DISCOVERY

One of the important lessons learned by the development team was the value of involving a group of individuals with a wide range of expertise and backgrounds as part of the project.

A major goal of *Project ScienceSpace* was to evaluate the medium of virtual reality for educational purposes. To properly do this required a considerable effort on the part of the developers in the proper analysis of their worlds. This included not only analysis during the design phase of each world as described above (e.g., the physics educators and focus groups), but also post-design experiments to compare the educational impact versus non-VR software.

While a majority of 107 physics educators felt that worlds such as *NewtonWorld* would be a beneficial means of presenting science to students, a direct comparison with existing educational software would be the real test. One area that offered a good place to perform this comparison was between *MaxwellWorld* and *EM Field™*, a “highly regarded and widely used” software product that also was designed to teach the concepts of electrostatic fields. In *EM Field™* (*EMF*) the user interacted with a two-dimensional representation of electrostatic fields. For the sake of comparison, the three-dimensional *MaxwellWorld* had features removed that were not part of the *EMF* software, leaving the dimensionality, physical immersiveness, and type (the quantitative *EMF* vs. the qualitative *MaxwellWorld*) of interface as the primary differences between the two.

By presenting the same physics lessons for groups using *EMF* or *MaxwellWorld*, the relative gains in understanding could be attributed to the choice of software. Both groups showed an increase in understanding, with *MaxwellWorld* users were better able to define concepts, and to demonstrate concepts in 3D, and yet still were able to sketch concepts in

2D at the same level as the users of *EMF*. Also, while users of the virtual reality interface were more likely to have difficulty with the interface, and/or experience simulator sickness, they still performed better than the *EMF*-using students. The developers believe that increased motivation in using the “new” VR system might be one factor that allowed students to overcome the experienced problems of VR.³

MaxwellWorld was used in an experiment to compare the effectiveness of egocentric and exocentric points of view on the virtual worlds. The designers expected that egocentric versus exocentric views would each be more amenable to learning particular concepts. They found this hypothesis to be true to a degree, but more importantly, they found that the mental task of integrating the global and local perspectives provided by the egocentric and exocentric points of view was a greater cause of learning because the user had to create a mental model that accommodated both.

The application developers made many other observations over the course of their work on this project, including:

- Each additional sense presented to the participant increased the participant engagement in the experience, and in general, increased learning. This assumes that the extra sensory inputs are designed to augment the primary sense, rather than pull attention away from the focal concepts.
- A single, long (e.g., 3-hour) VR session is generally not enough to improve the student's mental model of the concepts. This insufficiency results both from fatigue and sickness from the extended periods in the VR interface, and from the need to initially learn how to operate in VR.
- Three-dimensional concepts presented, learned, and tested entirely using 2D representations are not largely understood as three-dimensional phenomena even by students performing A-level work. “A” students in a high school physics class were good at

³The authors note that while the motivation factor may decrease as VR becomes more commonplace, this could be offset by less problems with the unfamiliar style of interface, and sickness induced by the limitations of the hardware.

giving correct descriptive answers in the 2D case, but they didn't have a good qualitative understanding of the concept, and were generally not able to answer questions related to the three-dimensional case.

Virtual reality allows younger students the opportunity to learn concepts that otherwise are taught much later in their educational experience because it can present them qualitatively, without relying on knowledge of higher mathematics. In particular, students in middle school could successfully learn the basics of Newtonian mechanics. This is particularly beneficial because this is approximately the age in which their childish mental models begin to give way to Aristotelian mental models. That is, the minds of these students are ready to create a formulaic mental model, so it would be better to provide them with a more appropriate representation that would help establish a good qualitative feel for the phenomena, and lead to a better, "intuitive" understanding of the higher mathematical descriptions. One of the potential pitfalls of using videogame-like technology to present scientific concepts is that kids have become accustomed to working in the nonrealistic worlds of many video games. Since this population is comfortable working within a nonrealistic world, they need to be convinced that the physics represented by the *ScienceSpace* worlds are indeed reasonably accurate representations of the real world.

The goal of *Project ScienceSpace* was to determine the potential of virtual reality for allowing students to gain better insights into scientific principles through a learning-by-doing approach. The multiple experiments performed using the various worlds of *ScienceSpace* have demonstrated that VR can be a useful tool toward this end. The developers saw that in the current method, in which students gather most of their information from lectures and reading, students who are able to perform well on tests may not in fact have a good understanding of how the concepts work in 3D. At the same time, the VR system allows the qualitative aspects of concepts to be taught at a much earlier age, before the point in the educational process when many students have already rejected the mathematics and hard science academic path.

Both during the design process and through surveys taken as part of the experiments, the *ScienceSpace* developers found that participants were much more likely to feel

a sense of presence, and in fact be more engaged in the educational activities, when additional sensory cues were provided. Whether this sense of immersion increases the ability to learn depends on how well the added senses augment the concepts being taught.

When students experienced difficulty with simulator sickness, or found it difficult to adapt to the virtual reality interface style of the experiences, their ability to learn and their sense of immersion were both diminished. Fortunately, only a few students experienced even moderate dizziness or slight nausea within the first 30 to 45 minutes, so by keeping VR sessions short, these problems can be managed. Most users were eventually able to get the hang of working in VR.

Overall, the *ScienceSpace* researchers believe immersion to be an important factor for allowing experiential learning to take place. As Chris Dede states:

Our research suggests that multisensory immersion for learning depends on actional, symbolic, and sensory factors. Inducing actional immersion involves empowering the participant in a virtual environment to initiate actions that have novel, intriguing consequences. For example, when a baby is learning to walk, the degree of concentration this activity creates in the child is extraordinary. Discovering new capabilities to shape one's environment is highly motivating and sharply focuses attention.

Related Applications and Follow-on Work

How important are foundational applications when planning new applications? *ScienceSpace* has inspired a new educational effort, to which we devote the entire following section.

Conclusion

Over the course of developing the *ScienceSpace* worlds, and testing their educational value, the project developers learned many interesting lessons.

Some lessons apply specifically to using virtual reality for elementary and high school education. They found that spreading lessons over multiple, shorter sessions gave students better success at assimilating and retaining concepts. They also discovered that virtual reality provided a mechanism whereby concepts that normally require abstract mathematical description could be taught in an intuitive qualitative

manner. By using three-dimensional representations, young children could learn the concepts even before acquiring the mathematical tools needed by traditional methods of instruction. However, the major problem with working with children younger than about 12 years old was that head-mounted displays are designed for adult-sized heads and are often too large for younger children, as was also seen with Strickland's work with autistic children (Chapter 5).

Some general observations about VR made by the project developers included experimental results demonstrating how adding feedback to more senses is important to the immersive experience. Higher degrees of immersion from multiple senses can improve knowledge acquisition as long as the added senses do not distract from the salient information. Another observation was that displaying the worlds in a projection-based display versus a head-mounted display results in a different method of interfacing with the worlds, although both were equally compelling.

Continuing work includes experiments designed to use *PaulingWorld* to study the value of VR for enabling students

“master counterintuitive chemistry concepts such as quantum-level phenomena.” *NewtonWorld* was adapted for a younger audience, and features were added that allow collaborative habitation of the world to investigate whether the nature of social constructivist learning enhances the educational value. Finally, *MaxwellWorld* was used to explore the “contribution of immersive frames of reference to understand complex science concepts.”

Project ScienceSpace has illustrated that virtual reality can provide benefits to science education. These benefits extend beyond the benefits already demonstrated by interactive computer applications such as EM Field, as revealed by the studies of the research team wherein the students were found to understand the world of electrostatics as a whole, better than the students using EM Field. [Dede et al.]

In the final analysis of course, it is not the fact that virtual reality was used in education that is important, but that education can be enhanced by virtual reality. Or, as the researchers stated: “The medium (virtual reality) should not detract from the message of learning scientific principles” [Dede et al.].

CASE STUDY 6.2: MAXWELLVU

Executive Summary

MaxwellVU is an educational application meant to augment the teaching of electromagnetics to electrical engineering undergraduate students and physics students. It concentrates on communicating the physical meaning of the fundamental Maxwell's equations to students in a virtual environment. Typically, education in electromagnetics is done through mathematical expressions and two-dimensional figures in textbooks, though time-varying media such as videos are sometimes also utilized. The revolutionary nature of the *MaxwellVU* suite of applications is to allow the student to step into electrical and magnetic fields, performing physical actions which change the fields based on their inputs. *MaxwellVU* is a suite consisting of a unique application for each of five equations,

allowing an educator to focus on the underlying meaning of each separate equation by allowing students to interact with simulations of electrical and magnetic fields. Students actively manipulate these virtual fields through hand and body motions.

The application is presently being used in teaching electromagnetics at Valparaiso University. It is available for download in two open-source formats. It has been ported and demonstrated in *CAVE* systems as well as a stripped-down desktop application for the PC.

Application Description

MaxwellVU is primarily concerned with giving students physical insights into the fundamental equations governing electromagnetics, the so-called Maxwell equations.

$$\oint_S \mathbf{D} \cdot d\mathbf{A} = Q_{r,S} \quad \text{Gauss's Law (Electric Field)} \quad (1)$$

$$\oint_S \mathbf{B} \cdot d\mathbf{A} = 0 \quad \text{Gauss's Law (Magnetic Field)} \quad (2)$$

$$\oint_{\partial S} \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial \Phi_{B,S}}{\partial t} \quad \text{Faraday's Law} \quad (3)$$

$$\oint_{\partial S} \mathbf{H} \cdot d\mathbf{l} = I_{r,S} - \frac{\partial \Phi_{D,S}}{\partial t} \quad \text{Ampere's Law} \quad (4)$$

as well as the related

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \quad \text{Coulomb's Law} \quad (5)$$

These equations are sure to appear to most readers intimidating at the least on first sight. Yet these equations comprise the mathematical expression of the laws governing electromagnetic field theory. Most physics undergraduates and nearly all electrical engineering undergraduates take a course in this subject.

Presently, undergraduate education in this subject is largely mathematical in nature. The physical meaning of the laws is touched on in lectures, and textbook material is augmented with diagrams (both 2D and rendered 3D). Additional media such as video exists on this subject, but is rarely integrated into an undergraduate course.

It is this present shortcoming in the pedagogy that motivates the use of VR to improve teaching methods. A further motivation is that electromagnetic fields are by nature invisible. VR offers the ability to render them visible, using different representations to show various aspects of their characteristics. Finally, these equations have real-world analogs, so that emulating the physical world in VR can lead to better understanding when the student finally has to tackle the math.

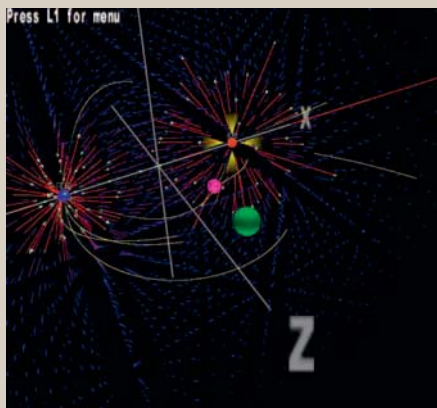
The *MaxwellVU* suite consists of five separate applications, one each to support understanding of each of equations (1)–(5) above.

VR DISCOVERY

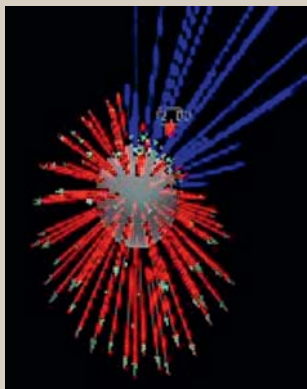
Who's afraid of the big, bad math? Whereas integral signs and Greek letters may terrify the average person, seeing loops of current and patterned arrows most likely does not. VR can be used to bridge the comprehension gap from mathematic expressions to inherent understanding. Immersing ourselves in the virtual world, we can have our hands become static electrical charges and watch magnetic fields change all around us as we wave our hands around.

The VR System

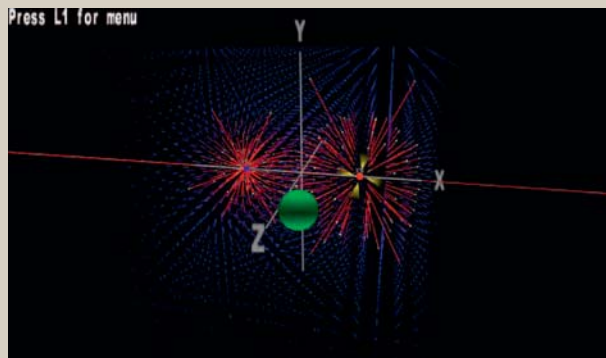
The system used in the five applications is a commercially available, turnkey system made by Visbox, Inc., Savoy, Illinois. The model is the Visbox-X2 system consisting of a $4\text{m} \times 3\text{m}$ rear-projected front screen and a $3\text{m} \times 2.3\text{m}$ top-projected floor. Button and joystick input is carried out on a handheld Wingman joystick device, consisting of eight buttons, two joysticks, a slider, and a D-pad. Head tracking is achieved through two front-mounted CCD cameras tracking two reflective spheres on the user's glasses. The system uses passive stereo through linearly polarized filters and glasses.

**FIGURE 6-8**

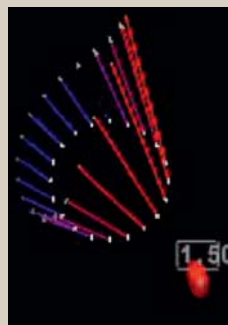
VectorVU allows the user to set up a constellation of point electrical charges and then shows the resulting electric field. Users may then move a cursor around with the wand and launch test charges that leave trails as they are attracted to negative charges. This enables students to understand both the characteristic fields generated from point electrical charges, as well as perceive the effect of the field on a charged free particle.

**FIGURE 6-9**

GaussVU allows users to move their hand as a “closed Gaussian surface,” reaching into and out of point electrical charges. The flux vectors are shown on the surface. The key comprehension imparted by this application is that the integration of the field vectors on the surface are proportional to the charge contained inside the volume.

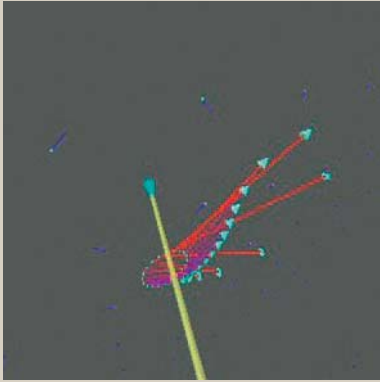
**FIGURE 6-10**

The ColumbVU application shows the electric field resulting from a negative and positive point charge. The green sphere is the cursor, while the yellow highlight on the red sphere indicated a selected charge. Students can place positive and negative charges within the 3D space, and view the resulting electric field.

**FIGURE 6-11**

The FaradayVU application, shows the electric field around a closed loop. Users can manipulate the electric field by moving static charges, yet the net sum of the electric field around the loop remains zero. Students see that in the absence of a changing magnetic field, the vector sum of the tangential components of the electric field must be zero.

The choice of linear polarization for the passive stereo puts a limitation on the orientation of the user's head. For the linear polarization to be effective, the vertical orientation of the user's head must be upright. If the user rolls their head

**FIGURE 6-12**

The AmpereVU application, showing the magnetic field along a closed loop. Shown is a circular loop, but other loops such as squares can be chosen. This application imparts to student the difference in the magnetic field around a closed loop when a current does and does not pierce the surface.

to the right or to the left, the polarization is lost. The use of circular polarization can eliminate this shortcoming, but suffers from higher cost and poorer filtering.

VR DISCOVERY

For a front-screen-only stereo display, linear polarization can be effective. Users tend to look straight-on to the screen and do not tilt their heads to the right or the left. This is typically not true in a 4 + -wall CAVE environment. When using just the front wall, SVL researchers find linear polarization effective. But the addition of a stereo floor caused users to have head motions and orientations that misaligned the polarization axes of their glasses. This resulted in loss of depth perception when users looked at the floor.

Interactions with the system are handled at two levels. When an instructor is using the system in a lecture demonstration format, students stand back from the screen such that their field of view is not covered entirely by the screen, much like watching television. The screen is projected in

**FIGURE 6-13**

Visbox-X2 system at Valparaiso University's Scientific Visualization Laboratory.

Photo by Aran Kessler

stereo and students wear polarized glasses. There is an area for the class to stand of approximately $5\text{ m} \times 4\text{ m}$, which can reasonably accommodate 20–25 students as well as the instructor. In this mode, the students are passive while the instructor holds the wand and controls the application while lecturing.

When students are using the system in hands-on mode for discovery learning, usually a pair of students utilizes the $3\text{ m} \times 2.3\text{ m}$ projected floor surface for interaction, in addition to the front wall. Standing on the projected floor allows the bringing of objects in front of the screen and provides for a more immersive experience.

The difference between the two modes of usage illustrates a valuable lesson when it comes to using VR for education. When students are passively watching a demonstration, the system is used much like a 3D viewing screen as opposed to an immersive experience. Even if the entire group is able to fit onto the floor surface, the crowded group essentially blocks the top-projection onto the floor. Thus the instructor tends to use the front screen only. Further, though, the objects represented in the virtual world are placed “behind” the screen since they cannot be brought forward without clipping the objects by the front screen’s extents.

This changes greatly when students participate in experiential learning using the application singly or with one other partner. Having the projected floor is essential for students bringing objects up close so that they interact with the surfaces, fields, and charges in close proximity to their own selves. The lesson learned: single front screens are great for presentations, but for student experiential learning, you need a floor.

Related Applications and Follow-on Work

ScienceSpace’s Maxwell’s World is the obvious forerunner of this educational effort. In fact, the chief developer of *MaxwellVU* was directly inspired by this foundational work. However, where *ScienceSpace* was limited to head-mount displays and was written for custom hardware, *MaxwellVU* uses open-source libraries and is able to be run on a variety of platforms.

This application is currently being used at Valparaiso University in their electromagnetic field theory class. A version of the software for the PC is available for free download, though the benefits of immersive VR are absent. The full-VR version is also available in open-source format

from the laboratory’s website. (Links are available on the companion website to this book.)

Instructor Dale Kempf, who utilizes the software in his class, says, “Electromagnetic field theory is a challenging subject for our students, partly because it is three spatial dimensions. Virtual Reality demonstration of Maxwell’s equations in three dimensions really help the students in my field theory classes understand the principles that are so difficult to visualize with two-dimensional media.”

The application is actively being used today, and undergoing further development. One planned improvement is the addition of time-varying magnetic fluxes that the users can arbitrarily manipulate with hand or body movements. This would give student greater insight into the time-varying aspects of electromagnetic fields. However, calculating resulting electric fields on the fly is complicated by the high computational requirements of solving a system of simultaneous partial differential equations. Presently, the bound is computational and limited by graphics capability. As CPU power increases, it is expected that this will soon be a surmountable problem.

Conclusion

Two aspects stand out from *MaxwellVU*. First, foundational applications such as *ScienceSpace* can motivate and inspire present application development, as is seen in this instance. Present VR development tools and libraries allowed the production of *MaxwellVU* in 6 months by a team of three, whereas *ScienceSpace* was in its time a near-Herculean effort. Other applications can build on such foundations.

Second, “tough stuff” like Maxwell’s equations can provide a perfect opportunity for the use of VR in education. Easing students into intimidating math by first exposing them to a virtual environment can both provide better class performance and lead to a better comprehension of the math itself.

CASE STUDY 6.3: VIRTUAL POMPEII

Virtual Pompeii premiered in the fall of 1995 (September 16 through January 7, 1996) at the M. H. de Young Memorial Museum of Art in San Francisco. *Virtual Pompeii* is a walk-through recreation of Pompeii as it existed before Mount

Vesuvius erupted in AD 79. Covered completely by volcanic ash, Pompeii, only a few miles from Naples, wasn’t rediscovered until the 16th century. The 5- to 8-meter thick layer of ash and volcanic debris plus other factors contributed to



FIGURE 6-14

The floor of the theater looking north. The stage is to the left, the beginning of the seating to the right.

this site being one of the best preserved of all archeological sites. This preservation aided in the ability to make a compelling recreation of the site.

According to Technical Director Jeffrey Jacobson, one advantage of doing a virtual recreation of the site is that it would be prohibitively expensive to create a real model. Plus, digital models have the added benefit that they can be easily replicated and distributed, even via the Internet.

By making the reconstruction available to a wider audience, it becomes more than a tool for archaeologists to envision the past. It becomes an avenue through which the general populace with an interest in human culture and history can experience the past in a way that is engaging and intuitive for non-archaeologists. The archaeologist is like an architect who can see in their mind how their plans will look when physically constructed, but whose clients are less experienced in cognitively converting the 2D plan diagram into a 3D mental image.

One of the goals of the *Virtual Pompeii* experience was to bring to life a more complete experience of what life was

like in Pompeii just before the eruption of Mt. Vesuvius. Rather than merely viewing photographs of an archeological dig side by side with a few artifacts from the dig and a drawing of how the structures were laid out, with *Virtual Pompeii* viewers can journey through the city and witness the society and culture “first hand.” When *Virtual Pompeii* opened at the de Young Museum, the application developers had not reached all of their goals, but did create a world that gives a flavor of what Pompeii was like in the days prior to its destruction.

Application Description

The focus of the virtual site is the Pompeian theater complex. The complex consists of the Large Theater, ruins of the Temple of Hercules,⁴ the Triangular Forum, and the Temple of Isis. These structures and other nearby streets and buildings are included in the *Virtual Pompeii* experience. Throughout the environment, animated figures (agents) add to the illusion of the resurrected Pompeii.

The *Virtual Pompeii* experience was designed to touch on the art, religion, and private life of the people of Pompeii. Participants can explore the walkways, sculptures, murals, courtyards, and structures as they existed long ago. The fact that Pompeii was nearly instantly buried served to produce a snapshot of the culture of the time. Often in other archaeological sites decay happens gradually, and new structures are built on top of the old, making it difficult to determine what things were like in any specific era.

Because the experience creators consider the work an educational application of VR, they paid considerable attention to the accuracy and detail of the recreation. The results were verified by scholars from the Archaeological Institute of America to be very authentic.

The temples provide a means by which the participant can wonder about the rites and rituals practiced in this mixed Greco/Roman/Mediterranean culture. For example, one might encounter the Temple of Isis. In the first century AD,

⁴The Temple of Hercules had been destroyed by an earthquake 80 years prior to the destruction of Pompeii. The structure had not been rebuilt by the people of Pompeii, so the VR experience left the temple grounds as a “picturesque ruin,” as it would have been in AD 79.

**FIGURE 6-15**

A chanting priest stands on the front platform of the temple.

Isis, the Egyptian goddess of nature, had a large cult following throughout the Greco-Roman world.

The Pompeian Temple of Isis is a small, ornate structure combining Egyptian, Greek, and Roman architectural features. The Temple grounds include the gardens and surrounding colonnaded walkway. Inside were separate structures for specific purposes. The focal structure was the sanctuary on a podium, adorned with sculptures. Inside the walls was a statue of Isis, which could be viewed through the front opening. At the front was a large porch where ceremonies could be carried out with the followers below.

A separate structure to the right of the entrance of the main building was a small roofless building containing a tank filled with water from the Nile River, mixed with rainwater. Behind the main building was a small garden area with a fountain and a statue of Bacchus (the Greek god of wine, also called Dionysus). The garden might have contained medicinal herbs as well as plants from the Nile, including at least one palm tree. The priests of Isis were famed for their medical knowledge. At the back of the surrounding colonnaded walkway were rooms believed to have been used for “ritual meals” and for “the mystery

**FIGURE 6-16**

The small building holding the sacred water, to the left of the main temple. A fire burns on an altar before it. Sacrifices of cloths, fruits, and other foodstuffs are shown burning on it.

cult’s dramatic initiation ceremonies.” The dining room contains a large marble table.

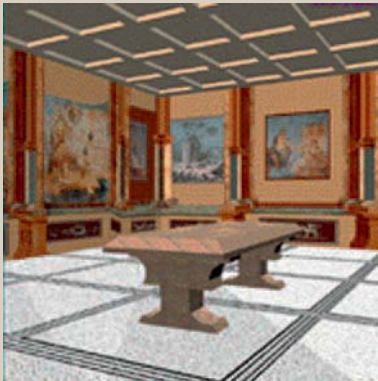
To help bring about a sense of how life was lived in the ancient city, the experience developers attempted to populate the virtual world with people, animals, and objects engaged in everyday life. Among the characters are a flautist, a chanter, and a group of singing birds.

Audio is used to enhance the sense of mental immersion. As the participant travels through the city, sounds such as singing birds, the melody of the flautist, and the voice of the chanter can be heard as the participant gets close to them. Many of the characters within the virtual world were implemented only as sound-producing agents. As the participant walked throughout the theater complex, there were several locations they could pass that would activate sounds such as a cat or a nest of birds.

Two agents that were fully implemented were a priest playing the Aulos (a double-barreled flute) from which music can be heard when the participant is nearby, and a priest singing an adoration of Isis. The singing priest stood on the portico of the Temple of Isis waving his hands out

**FIGURE 6-17**

This image depicts the view from the ceremonial dining chamber at the back of the temple complex. The back of the temple is barely visible to the right and can be seen just to the right of the right-most column.

**FIGURE 6-18**

Another view of the ceremonial chamber, showing the murals.

and up in the manner of Roman prayer. The flautist was also within the grounds of the Temple of Isis.

There were several places in the complex that would transform words from classical Latin to English as one approached the subject. For instance, from a distance the chanting priest could be heard in Latin, but up close, the words switched to English so the listener could better understand the words.

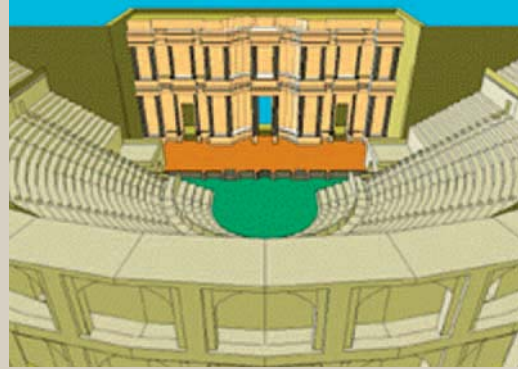
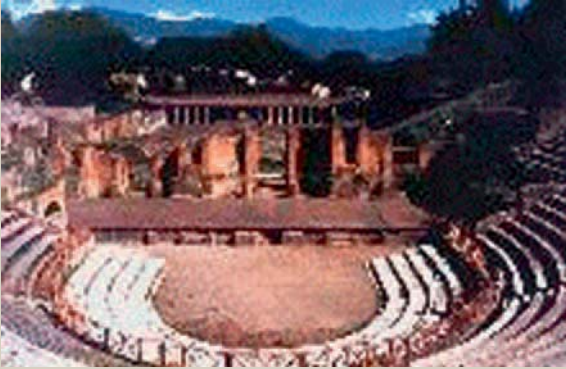
**FIGURE 6-19**

The Grand Theater (as modeled by Veronica Polo).

Both versions of the chant were sung by the same person with the same tempo and inflections, so the transition was remarkably smooth. A similar effect was done with the text on plaques, which can be found throughout the complex. As with the voices, from a distance, the words appear in Latin, but as the participant approaches they transform into English.

Constructed between 89 and 75 BC, the Pompeian amphitheater is the oldest known amphitheater. It could hold 20,000 spectators, nearly the entire population of the city. One of the more interesting aspects of the virtual reconstruction is a virtual re-creation of a theatrical performance in the Large Theater located in the center of the large open-air amphitheater. The play automatically begins as the user enters the amphitheater. The Carnegie Mellon University Drama Department and the University of Pittsburgh History of Art and Architecture Department helped to provide the content of the play. While in the auditorium, the participant hears the actors and the accompanying music.

The play itself was written at Carnegie Mellon and produced by Professor Don Damaellio. At the time of the eruption, Roman theater was very slapstick, with vulgar humor presented in a vaudeville-like style. Since the content of

**FIGURE 6-20**

The Pompeian amphitheater is the oldest known amphitheater. Shown here are both a picture of the actual excavated temple and its corresponding virtual model.

actual plays from the period would be offensive to the modern museum-going audiences, it was decided to create a new play in a similar vein, but tuned to modern audiences. Roman historian and archaeologist Jane Vadal contributed many suggestions and other input to the development of the play. The result was a political satire very much in the vein of the then popular fare, but involving modern-era issues.

The content of the play was a combination slapstick and political satire about the Emperor Toitas's older Egyptian mistress. Given the opportunity to incorporate a new play into the experience, Vadal felt that it would be better to use a portion of a play by Plautus. By AD 79, Plautus's works were old and no longer fashionable, but were still performed in provincial places like Pompeii. More importantly, according to Vadal, they were slapstick masterpieces.

In addition to changing the content of the play to more closely match the expectations of the modern viewer, the developers also changed the coloring of many of the buildings for a similar reason. Because the Romans liked very bright, garish colors, sticking to the authentic colors would have been distracting to the modern eye. So in effect, these changes were made to allow participants to see the world closer to the way a Pompeian would have seen the world than if the world were duplicated more exactly. A Pompeian would not have thought twice about the color of the buildings or the content of the play, whereas a modern viewer would be more distracted by these particular elements of the world.

This technique could be classified as “perceptual cue tuning,” where changes are made to how the world is presented to make the world appear more natural, even though it has to be somewhat distorted to make that happen. For example, the original colors of the buildings were toned down to accommodate modern sensibilities.⁵

The play was created basically as a 3D animation that can be viewed from any angle. The movements of individual performers were recorded using typical motion capture methods. These performances were then combined and choreographed together with the help of the Jack™ system from the University of Pennsylvania. This was a very painstaking process, and in the end the look was not as good as the developers had hoped, so the visual portion of the play was left out of the final experience, presenting only the audio to the participant. The only method of interface with the world is via the travel controls. A trackball (2DOF input device) is used to move the user through the world forward, backward, or laterally. For most venues, a physically immersive HMD display is worn by one participant. This person is also given control of the trackball to move through the city. For larger, theater-like venues the *Virtual Pompeii* experience could more easily be presented on projection screens by following a preprogrammed ride-along path.

⁵See the “Officer of the Deck” application of Section 1, Chapter 7 for more discussion on perceptual cue tuning.

VR System

The *Virtual Pompeii* was designed to operate at regional culture centers. The first venue for a planned 2-year tour beginning in 1995 was the M. H. de Young Memorial Museum of Art in San Francisco. Unfortunately, the planned tour didn't happen, and a possible presentation at the Smithsonian Institute for the summer of 1998 also fell through. After the experience closed at the de Young Museum, *Virtual Pompeii* was presented once at the Carnegie Science Center's ElectricHorizon™ interactive theater and in many private showings at CMU.

At the de Young Memorial Museum, the exhibit was arranged such that an attendee's experience begins in the hallway leading up to a room that had been set up as a mini-theater where the *Virtual Pompeii* was housed. The mini-theater could hold about 40 people. Entry to the *Virtual Pompeii* exhibit was included in the regular admission price for the museum.

Along the entryway were real photographs of the Italian ruins used to reconstruct the ancient world, replete with explanatory text. Upon entry to the theater, one person was chosen to wear the HMD. This person was generally chosen on a first-come basis. A member of the museum staff assisted this person with the equipment and provided an informative narration. Others watched the experience vicariously on overhead screens. Both the HMD and overhead screens could be displayed stereoscopically or monoscopically. When a stereoscopic image was displayed on the large screens, polarized glasses were given to the viewers to allow them to properly see the image.

In prior experience with presenting re-created ancient Egyptian structures in a public museum venue, the application developers found that the audience could be possessive about how much time they should be allowed to remain immersed in the world. Based on this experience the developers decided to implement a clear and dramatic ending to *Virtual Pompeii*. Fortunately, for this particular scenario, there was an obvious action available that would end the experience within the context of the experience—the eruption of Mt. Vesuvius.

As the visitor's time expired, they would first hear a recording of the rumble of an actual volcano. A few seconds later,

the world would begin to shake, represented visually by perturbing the viewpoint. At first, 3D particles spewing from the volcano began to occupy more and more of the participant's visual perception. After a few more seconds, the entire scene begins fading to a snowy grey image. Finally, as the rumbling sounds continue in the background, the visitor's view becomes entirely filled with ashen grey.

Just as abruptly as the lives and culture of Pompeii ceased nearly 2000 years ago, the participant's experience in the virtual world ends as they get buried in ash. Although the eruption of Mt. Vesuvius was not as instantaneous as many laypeople believe, this belief could be used as an excellent narrative device for ending the experience quickly. The entire sequence from first hearing the audio rumbling to the completely grey scene occurs in 20 seconds.

The application had controls to provide a 3-, 5-, or 10-minute experience prior to the eruption of Vesuvius. Once all vision turned ashen grey, the participant removed the HMD, and the next person entered Pompeii. The different settings for the length of the experience allowed the operators of the venue to adjust the throughput based on the size of the audience that came for the experience.

This project utilized two Silicon Graphics Indigos™ and one Onyx™ for development. A second Onyx, which included four processors (R4400, 200MHz), and a single RealityEngine2™ graphics engine were used for the traveling display. The Onyx was capable of supplying up to 30fps at 640 × 480 resolution. Head-mounted displays from Liquid Image and N-Vision were used to display the visual images to the head-tracked participants. The head tracking was done with an Ascension Technologies electromagnetic tracking system. Large screen projection was done using Electrohome Marque projectors.

The higher-resolution n-Vision HMD (640 × 480 pixels) was deemed to be too expensive and fragile for the public show, so its usage was reserved for private demonstrations at the CMU lab. The developers felt that the Liquid Image HMD, although lower in resolution (256 × 256 pixels) was a better solution for the museum display as it was “physically able to withstand the beating it got from being in a museum show.”

Similarly, a third-party trackball was used as the input device because it was sturdier than other options. The trackball also seemed to be an interface that would be more

intuitive for controlling walkthrough travel, where the user can move along the surface in two dimensions.

Silicon Graphics *Performer*TM software was used for the image generation. In addition to rendering the images, *Performer* provides capabilities for handling the collision detection, database management, and stereo image synchronization. Modeling was done with the SIMLAB's preferred packages, *form-Z*TM from Auto-des-sys, Inc. The choreography was done with JACKTM from the University of Pennsylvania Center for Human Modeling and Animation.⁶

Other products used during development of the application include tools from VSI Visual Synthesis, Inc. to render the audio; Adaptive Optics Associates for motion capture; and Corphaeus Inc. and Lightscape Technologies for additional graphics support. As with the HMDs, some of the advanced features provided by some software tools were not at the de Young showing.

The creation of the virtual world model was very difficult. Roman scholars from the Archeological Institute of America (AIA) were assigned to work with Jane Vadal, the project's Research Director. She took the information they gave her, along with data from other sources, and specified the reconstruction of the architectural details, murals, and other artwork. All of this was derived from sources such as early German watercolor paintings and black-and-white photographs. Text descriptions from historians such as Pliny who lived during the time of the eruption were also used to add details to the world. Computer modeling was done using the *Form-Z* package, and later, *Designer's Workbench*. The actual (digital) reconstruction was performed by the art team.

Virtual Pompeii was a joint project between The Archeological Institute of America, Silicon Graphics Inc. and the STUDIO for Creative Inquiry at CMU. The team was led by Carl Loeffler, who was the research director of SIMLAB, the STUDIO's virtual reality research team. SIMLAB member Jeffrey Jacobson served as technical director on the project. It was created by programmers, artists, sound engineers, and musicians from CMU using the STUDIO for

Creative Inquiry facilities. The Archeological Institute of America provided consultants to ensure accuracy in the areas of architecture, gardens, painting, and sculpture. The Drama Department at CMU provided content specialists for Roman Theater, and the Department of Art History provided content expertise in Roman Art.

Related Applications and Follow-on Work

In a related application that allows people to walk through Pompeii, scientists at Max Plank institute have created a "CyberCarpet" to enable visitors to a virtual creation of Pompeii to *physically* walk through the world. The CyberCarpet is an omnidirectional treadmill that allows a participant to walk endlessly in any direction. The participants wear a head-mounted display with markers on it for optical tracking. The virtual Pompeii was implemented with CityEngine, which allows creation of large-scale virtual environments. CityEngine was created by the Swiss Federal Institute of Technology (ETHZ).

Conclusion

The elements that were implemented demonstrate the possibilities for using VR as a tool for advanced and casual study of lost cultures. For the casual observer, it affords them the opportunity to imagine Pompeian life without reading a detailed book on the subject. For the serious student of archaeology or anthropology, it provides a tool to create a mental model of the world much more quickly than possible by pouring over photos, sketches, and field reports.

One of the benefits of putting site reconstruction through the rigors of creating a 3D virtual world is that the 3D reconstruction is less able to support contradictions and gaps in the data interpretation than other media. In particular, a written description or a mental model based on incomplete and unclear references are both able to accommodate gaps and contradictions without being noticed. Thus VR forces archaeologists working on a site reconstruction to confront opposing and missing data.

The *Virtual Pompeii* experience proved itself as a worthwhile effort at addressing an archaeological site reconstruction, and medium for public presentation to casual observers. During the exhibit at the de Young Memorial Museum, there was a marked increase in attendance due to the VR presentation. With a background in both art and computers, Renée Dreyfus, curator of the museum,

⁶And later released as a product from Transom Technologies, Inc., a subsidiary of Engineering Animation Inc. (EAI).

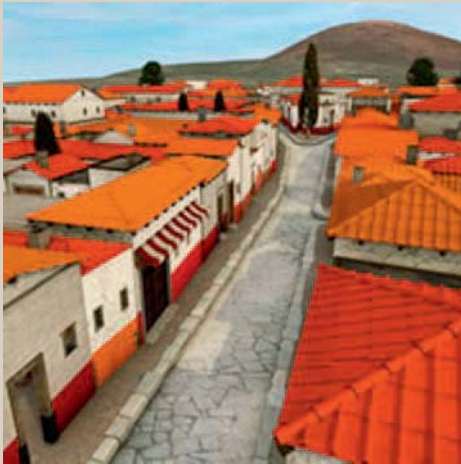


FIGURE 6-21

A virtual reality walkthrough of Pompeii was created using CityEngine software from the Swiss Federal Institute of Technology. The image on the right shows a participant walking on an omnidirectional treadmill to experience Pompeii.

showed great excitement about this new type of interactive exhibit. She stated: "I'm interested in seeing how far a museum can use technology for presenting contexts for works of art, and helping the public to understand the art on display. What better way than to bring them back to a world that no longer exists, but can exist in a computer?" Jacobson discusses how VR can be used to create an experience that allows a participant to experience another (no longer extant) culture:

These reconstructions open the door, not just to the buildings, but to the life that the people lived back then. We can have a sense of people going about their business. In a well-researched project, viewers will experience the weather as well as the natural flora and fauna of the time. The viewer of an historical reconstruction has a rich visual synthesis that would take years and years of study to otherwise build from books and artifacts.

CASE STUDY 6.4: VIRTUAL REALITY GORILLA EXHIBIT

Few virtual reality applications have focused on teaching about the social behaviors of the animal kingdom. The application described here is remarkable in that it realistically portrays the behavior and social interactions of a community of gorillas. The application allows a participant to *become* a member of the community. As an adolescent gorilla, the user can interact with other (virtual) gorillas to learn first-hand about the pecking order, and rituals of a gorilla society.

The *Virtual Reality Gorilla Exhibit* was a joint project between Georgia Institute of Technology's Graphics, Visualization, and Usability Lab (GVU) and Zoo Atlanta. The project had an overall goal of educating middle school children about gorilla interaction, vocalizations, and social structures. Because gorillas are an endangered species, gorilla conservation education is becoming increasingly important. The project designers felt that by allowing students to



FIGURE 6-22

Students visiting Zoo Atlanta are given the opportunity to enter a gorilla community to study gorilla social structures. In the image on the left, a student dons a head-mounted display and holds a wand. A ring platform provides a boundary for the student. Note the actual gorilla visible through the window near the VR exhibit. The image on the right shows a virtual recreation of the Zoo Atlanta gorilla exhibit that the student sees.

Photo by J. Sebo/Zoo Atlanta-copyright 1996



“become a gorilla” they might increase understanding and concern about the issues facing gorillas.

The developers (GVU researchers in conjunction with zoo personnel) established two separate educational goals. The original goal was for middle school children to learn about dominance in a hierarchy of gorilla social interactions. Later, a new goal of teaching zoo habitat design philosophy was implemented using the same underlying software. It was hoped that by enabling students to explore within the context of the gorilla society they would gain a better perspective of that society. The habitat design application provides a means for studying the psychology and philosophy of environmental design at the college level. For both projects, the novelty of VR was thought to be helpful in that it might increase the students’ desire to explore the virtual world. The application of VR to middle school zoological education is what is focused on in this section. The dissimilarities between the applications supporting the two goals will be briefly discussed toward the end.

Prior to the *Virtual Reality Gorilla Exhibit* project, students could only learn about many aspects of gorilla behavior via reading second- and third-hand accounts. Direct observation of the gorillas in the zoo does allow some opportunity for study, but it is inadequate in many ways. For example, gorillas are active early in the morning and late in the afternoon, but are asleep during the early afternoon when the students would typically have the opportunity to observe. Additionally, because many parts of the zoo are inaccessible to the general public, the VR exhibit enables students to investigate many otherwise unseen aspects of gorilla habitats. The introduction of new gorillas to the community is not observable, because it is done outside the view of the general public; thus, students are unable to observe the acclimation process of a new gorilla. In general there is no way a student can join with the gorillas and interact with them as a peer. The *Virtual Reality Gorilla Exhibit* addresses these issues, and allows a student to have a firsthand learning experience related to the life of a gorilla and its habitat.



FIGURE 6-23

A variety of gorilla postures were modeled for the Virtual Reality Gorilla Exhibit. The image on the left shows a male silverback striking a pose, and the image on the right shows its virtual counterpart.

Image courtesy Don Allison

Application Description

The application designers endeavored to create as realistic a world as feasible, while retaining a relatively high visual frame rate. The application includes a model of the gorilla habitat (modeled geometrically) and gorilla agents (modeled geometrically, behaviorally, and sonically). Real-world images and sounds were incorporated into the experience as texture maps and sound samples to enhance the realism of how the world is presented.

The gorilla behavior model was created based on text and video accounts of the gorillas. Local zoo experts helped refine the model. Five gorilla models were constructed: an adult male silverback, an adult male blackback, an adult female, a juvenile, and an infant. The gorillas were created as three-dimensional entities modeled to look and act like actual gorillas. The gorilla motions were created by modeling a series of key poses. Intermediate positions are generated by interpolating between poses. When video of a pose or motion was not available for modelers to study, zoo experts would act out the role.

Each gorilla motion has an associated vocalization. The vocalization rendering is timed to correlate with the motion. More than 80 gorilla sound samples were collected

for the experience. From these samples, a few general classes of sounds were created: an ambient sound of running water (from a waterfall in the habitat), mixes of male/female contentment sounds, coughing/annoyance sounds, a generic roar/charge/chestbeat sequence, and a separate chestbeat sound.

In addition to accurate portrayal of the gorilla family, a model was created of the gorilla habitat at Zoo Atlanta, including such details as the foliage, trees, and rocks. Topographical data was used to create the terrain. The creators also modeled the two associated buildings, the Cameroon Interpretive Center and the exterior of the night holding structure (they have not had the opportunity to model the interior). The Cameroon Interpretive Center is a building at Zoo Atlanta with large glass windows through which zoo visitors can observe a portion of the habitat housing a silverback gorilla (Willie B) and his family (several female and juvenile gorillas). This exhibit (called the "Gorillas of Cameroon") is also a focal point for educational activities.

As an educational application, there must be more to the experience than wandering around the habitat and watching the gorillas. The subtleties of gorilla movement

**FIGURE 6-24**

The Virtual Reality Gorilla Exhibit modeled the actual Cameroon Interpretive Center. These images show the actual center (the building in which the VR display is housed) and the VR model of the facility. In the gorilla experience, the students begin in the virtual center, and step through the window into the gorilla pen.

Image courtesy Don Allison

and postures can be better learned when an explanation of what is being witnessed is presented along with the view of the world. In this case, such a presentation would require both that the gorillas act in a natural fashion and that a narrator agent or other presentation mechanism be programmed to give context-sensitive information. The initial *Virtual Reality Gorilla Exhibit* included two agent gorillas with which the participant could interact with an adult male silverback and an adult female. For the initial public presentation, these tasks were performed by zoo experts, and not the application itself.

VR System

An untracked, handheld prop equipped with two buttons provides the user input interface. Travel about the environment is via a gaze-directed walkthrough method. Direction of travel is based on the direction the participant is looking. There is no means of independently rotating the world, so a full horizontal field-of-regard (FOR) display is necessary to guarantee the user will be able to always see where they are going. Thus, this interface technique works best with HMDs and projection systems that completely surround the viewer. The buttons on the hand controller activate movement in the longitudinal (gaze) direction at a constant rate of travel either forward or backward depending on which button is pressed. In the habitat design application (described later), a 6-DOF tracked stylus with

a single button is used instead as the handheld input device. The tracked stylus allows for a point-to-fly method of travel plus other forms of interaction needed for doing design work.

To enable this experience to run on lower-cost computers, some rendering tricks were necessary to maintain a reasonable frame rate. Because movement in the virtual world was constrained to areas in which the gorillas were free to roam, there were certain facets of the space that could not be seen. The developers used this constraint to enhance performance by removing unviewable polygons. Polygon count of the terrain was also reduced wherever possible. To maintain the basic shape of the terrain, some criteria were set for polygon removal. Polygon removal should not change the terrain slope by more than 5 degrees per 2-foot interval within areas the user could explore. In areas that the user could see, but not enter, the criterion was 10 degrees per 2-foot interval. The rendering system needed also to be capable of hardware texture mapping to allow all the polygons to make use of textures to enhance the scene realism.

Students experience the virtual world via a head-mounted display that provides a monoscopic image displayed to both eyes (the monoscopic-binocular display method). A student stands inside a ring platform with a railing that helps keep

them within range of the cables and tracking system. The ring also provides a source of stability to aid the students in feeling assured that they will not fall down or walk into anything.

The initial application was developed on an SGI Onyx computer with an RE2 graphics accelerator. They ran the gorilla applications on either a Silicon Graphics Indigo 2 with Maximum-Impact graphics, or a “medium- to high-end PC” with a Pentium II™ Intel processor in conjunction with a hardware OpenGL accelerator card. The Silicon Graphics systems provided a better frame rate than the PC-based system (about 15fps vs. roughly 10fps), but at a higher cost.

Full 6-DOF position tracking (both location and orientation) was done of the participant’s head. No other body part was spatially tracked. The sound is monophonic and does not vary according to head position (it is presented with a head-referenced sound stage). Sound display was done via headphones in the HMD and through a subwoofer positioned beneath the ring platform. The headphones were later replaced with loudspeakers both to allow others to hear the sounds of the experience and to allow the participant to converse with the others watching nearby. The software environment used to create the world was Georgia Tech’s Simple Virtual Environment Toolkit (SVE) [GVU Tech report 93–13].

A vicarious viewing display system was also provided for the press, public, and other students to view and hear the action. Two large monitors presented the participant’s view, and speakers in the pavilion presented the sounds.

Trial Run

Once a prototype of the *Virtual Reality Gorilla Exhibit* was ready, an informal study of the VR experience was held. Participants were children aged 7 to 15, most of who were already involved in the zoo’s Young Scientists program.⁷

Known as “the Gorilla Squad,” this program gave children the opportunity to observe gorilla behavior, so most of them were already versed in watching the gorillas though from outside the habitat. The VR system was installed in the Gorillas of Cameroon center and a day was spent (amidst press coverage) allowing the students to try the VR experience. Each session began with the student participant exploring the virtual “Gorillas of Cameroon” exhibit. About 30 people had the opportunity to experience the world first hand.

The VR platform was located next to the window overlooking the gorilla habitat in the Zoo. At the start of their experience each student was placed within the (virtual) center, viewing the (virtual) gorillas through those same (virtual) windows. After getting familiar with the HMD and navigating the world by exploring the center, students were instructed to walk through the windows and into the gorilla habitat. As they passed through the window into the habitat, they were told that they were now juvenile gorillas.

Students were free to explore the environment, by “walking” around, and perhaps making attempts to initiate contact with the virtual gorillas, which would react appropriately to their actions. The gorillas in the virtual world were implemented as agents programmed with many gorilla behaviors. However, for the prototype version of the application, only lower-level behaviors were preprogrammed. High-level gorilla behavior was implemented using the “Wizard of Oz” technique. A zoo expert controlled the high-level behavior of the gorillas using a keyboard interface. This technique enabled the gorilla family to react appropriately to the student actions. The keyboard interface has keys for selecting which gorilla was under “wizard” control, and other keys to give commands to the current gorilla, such as standing, walking, or making a bluff charge. The gorillas not under “wizard” control continue performing the last request given to them. There are also key commands for administrative operations such as resetting the positions of the user and all the gorillas.

By incorporating lower level behaviors into the model, the “wizard” can instruct the gorilla to perform actions such as “move to point A” without having to specify every muscle or joint movement. As the virtual gorilla performed this instruction it used programmed “reflexes” to avoid obstacles such as holes or trees. The “wizard” could select

⁷A program of the zoo that already existed in which school children come to the zoo on a weekly basis for 6 weeks to observe gorilla behaviors and take ethological data.

from many common poses, which were presented with the appropriate gorilla vocalization.

In the real world, gorilla communication is by vocalization, posture, and facial expressions. Some expressions of displeasure may be perceived by human observers as physical threats. However, in the virtual world participants know that their physical well-being is not actually being threatened. This can lead to participants' actively seeking out and performing actions that annoy the adult gorillas, to provoke a threatening response. Because this goes against the educational goal of learning how to behave as a proper adolescent gorilla, an alternate form of discouragement is incorporated in the VR experience. This discouragement comes in the form of reducing the interaction with the VR experience. They are put in "time out," where they are unable to see the gorilla environment, only a message reminding them that they are being punished.

First Impressions

Overall response to the experience was positive. Students felt the experience was "fun," and that they did learn about gorilla behavior. Zoo observers saw that participants acted naively at first, but later recognized and responded appropriately to gorilla cues and signals. Most of the users interacted as they would in the real environment. For example, they did not jump into a 12-foot ditch that was there even though they were not prevented from doing so. The developers felt that the students achieved a high degree of immersion, and that there was transference of knowledge gained in the virtual world back to the real world.

Some observations made at the experience trial are similar to observations of other VR experiences at other public venues. Younger students spent more time exploring the environment, whereas older ones spent more time observing and interacting with other gorillas. Most students did not make many physical movements. Typically, they would stand stationary in the platform ring, occasionally rotating their body to look and move in another direction. However, they did travel extensively throughout the virtual habitat. Students expressed that they wanted to interact with other "student gorillas" and to see their own gorilla body. The students were disappointed they couldn't touch and feel the other gorillas, or feel the fur of their own body. Because the experience designers believe current haptic feedback

devices are not yet capable of providing acceptable representations of this nature, they plan to design future efforts to minimize the need for such devices.

One confusing aspect of the initial virtual world was that the audio display of gorilla sounds was of constant amplitude regardless of how far the student was from the source of the sound. Thus, if a distant gorilla made a sound, the student perceived it to be very close and would look for it, expecting it to be nearby. They did feel the sound was an important component, nonetheless. The problem with the audio cues was changed in later versions by adjusting the amplitude of each sound based on the distance from the viewer. It was still done monaurally, so only distance audio cues were provided, without directional cues.

Further Development

From the prototype experience, the developers found that the experience was much better when participants were "accompanied" by a knowledgeable guide who could point things out and otherwise provide assistance. However, it is not practical for the experience to require a crew of experts to act as assistants and Wizard of Oz agents. Therefore, the developers pursued an enhanced behavioral model for the gorillas capable of performing realistic high-level interactions. They also investigated integrating some form of "automatic" guide.

A significant enhancement was to further automate gorilla behavior using the same low-level poses and actions available to the zoologist running the Wizard of Oz interface, plus a built-in annotation mode that would elucidate that behavior. These behaviors were programmed with input from Zoo Atlanta zoologists. An additional significant benefit enabled by automating behaviors is that the behaviors of each gorilla can be computed in parallel. Thus, multiple gorillas can react to the participant, or even to each other.

When the Wizard of Oz approach was used, the "wizard" could only specify the reaction of one gorilla at a time. Therefore, they would always concentrate on the interaction between the participant and the gorilla nearest the participant. However, with gorillas now able to interact with one another automatically, the participant might set off a chain reaction by causing one gorilla to move, and thus change its relationship with the other gorillas, causing them to react accordingly. For example, if the participant invades the space

**FIGURE 6-25**

Colored shapes indicate the mood of individual gorillas.



of the silverback, inciting it to charge, the silverback may end up nearer to one of the adult females who had been sitting peacefully. The silverback may then indicate to the female that she is too close, in turn resulting in the female becoming submissive and moving away from the silverback.

Personal experience is a powerful way to learn, and is one reason why people are exploring virtual reality as a teaching tool. Bowman and Hodges pointed out, however, that “students may learn by experience, but they may also be confused or draw incorrect conclusions” [Bowman and Hodges, LJVR]. To help make the gorilla society understandable to the participant, someone, some thing, or some autonomous agent must provide an interpretation of the gorillas’ postures, sounds and reactions. An initial thought was to develop an automatic guide agent that would act very much like the zoologist making comments over the shoulder of the participant. However, they later decided to use an annotation system as the primary means of presenting this information.

There were three types of annotation presented. First, a symbolic mood indicator could be shown above the head of each gorilla. Gorillas could be in one of four moods:

- Content and docile—represented as a green cube
- Mildly annoyed—represented as a yellow upsidedown triangle

- Very annoyed—represented as a red octagon
- Submissive—represented as a white pennant

The symbols chosen to represent these moods are obviously based on symbols ingrained in the target users as members of modern Western culture such as a yield sign, stop sign, or white surrender flag.

As a junior member of the society, the participant cannot cause any of the gorillas to become submissive to them, and could only witness this mood when two other gorillas interacted.

The second form of annotation is presented aurally, in much the same fashion as would be presented by a live expert companion. These annotations are presented when a gorilla’s mood changes and the student doesn’t know how to interpret the expressions of the gorilla. For example, mildly annoyed gorillas might make a coughing noise and look away from the offending gorilla. As a novice to gorilla society, the student will need to be told what this means. In this case, an audio annotation will state: “Coughing and gaze aversion are signs that this gorilla is becoming annoyed because you are too close to him. You should quickly look away and move away.”

The last type of annotation provides instructions on how to interact with the virtual reality system. For instance,

users will be told how to travel; once they have attained a basic ability, they are instructed to fly through the window, into the habitat, and are told that they are junior members of the gorilla society. These annotations are also provided vocally.

Each type of annotation could be disabled. Thus, the system could be used to test students after they had the opportunity to explore it with full annotations. Typically, the student would explore the space with full annotations until they became comfortable with it, then all the annotations could be disabled. The audio and iconic annotations can be enabled and disabled separately. Thus, students could continue to use one information source after discarding the other. Most audio annotations were programmed to be heard only the first time the participant encountered the corresponding object or event, such as the first time they climb on a dead tree. In this case, they will hear audio explaining why the trees are there, but they won't hear that message again even when they climb on a different dead tree.

A Related Effort—Habitat Design

As mentioned earlier, a separate educational experience was created using the same virtual world as the societal experience. This new version of the application was for a different audience. Rather than focusing on introductory science exploration for middle school students, the new application focused on the study of design philosophy for college architecture students.

As opposed to many other VR examples of design applications that provide the user with a blank canvas, the developers determined that it was difficult to develop a world from scratch using VR. Thus they decided to provide an existing space, and allow students to modify the design. The beginning layout was the same as the main gorilla habitat at Zoo Atlanta. This was the same world used in the society application.

Students were presented with a variety of ways to alter the space. These included moving objects such as trees, rocks, and foliage barriers. The moat surrounding the terrain could not be modified, but the terrain itself could be drastically altered. However, these drastic alterations were all predetermined based on standard zoo habitat design philosophies. For example, some zoo designers prefer to

use a “bowl” design to allow visitors to more easily gaze into the area, and see the animals no matter where they are located. Another design philosophy is to use steep hills, allowing the animals to hide from view if they choose.

The application designers strove to use good design practice in creating the interface and information representation. Changes were made from the original application regarding the method of travel control, the ability to select and manipulate objects, and the amount and type of information annotations.

In the societal application, there was no opportunity for selecting or manipulating objects in the world. Thus, a new interface was necessary to allow design alterations. One way to move objects is to select them by making contact between the prop avatar and the object, and then pressing a button to move the selected object along with the prop. To allow objects to be manipulated at a distance, the representation of the prop moved using a “go-go” arm technique. This technique provides a mechanism by which movement close to the body maps directly to movement in the virtual world, but as the arm is extended, the distance covered in the virtual world increases exponentially, allowing the user to reach very far. This is essentially a drone-contact method of selection/movement, and because the prop avatar isn't always co-located with the physical prop may be confusing in a nonocclusive display (this works best in an occlusive HMD).

An alternate method for moving objects is to use a stylus and tablet interface. In this method, the user holds a prop tablet in their off-hand, and a pen or stylus in their dominant hand. By raising the tablet into view, they can see a map and menu representation on the virtual tablet. The stylus is then used to point to locations on the map, or menu options. Thus, by pointing to an object's iconic representation on the map (e.g., a triangle representing a tree), the user can press a button and slide the tree to another location in the virtual world. As the tree is moved on the map, it can also be seen moving in the 3D world.

There is also a choice of how to travel through the world. Rather than using the gaze-directed travel interface, the handheld prop can now be used to indicate the direction of travel. This allows the user to move laterally, while



FIGURE 6-26

A handheld, tracked tablet is used as a user interface in the habitat design application.

enabling them to maintain visual contact with a particular part of the world as they move. Pointing up allows the user to fly. As a gorilla in the society application, the user is constrained to walk on the ground, but as a designer, there is often a need to move back for a global view. However, the user is still constrained to remain above ground and within the horizontal confines of the habitat.

A second form of travel is available using the 2D map on the tablet. In addition to representing the location of objects in the habitat on the map, the user's location is also indicated (with a red dot). Just as moving the object icons results in movement of the corresponding objects, moving the user icon moves the user to the specified location in the habitat. The one difference is that the user movement doesn't take place until after the icon is no longer being manipulated. This delay in movement allows the user to maintain a better sense of navigational awareness.

To allow design students the ability to learn about and access the habitat, several annotations are located throughout the space. Annotations can be associated with objects (e.g., a tree) or with locations (e.g., the visitor center). Annotations can be vocal, text billboards, or pictorial billboards. Vocal annotations also have a visual representation indicating the topic of the annotation, and they can be programmed to be played only the first time they are

encountered, or every time. The user has some control over the types of annotations that are visible in the space.

The application developers conducted two disparate studies using this application. Both studies addressed the usability of VR for education, one as a general test of the usefulness of VR [Bowman, Wineman and Hodges], the other as a specific usability analysis of the effectiveness of this application's interface [Bowman, Wineman, Hodges and Allison].

In the first experiment, some members of a class on design psychology were told they were participating in tests of new technology that might be used for the course in the future. In fact, the class was split into three groups: a control group that did not have the chance to use the system, a walkthrough group that could only wander around the static environment, and a group that could walk around, and also access the informational annotations. A few days later, the class took a test with questions that could be answered based on class lectures only, the VR experience only, or both. The sample size was small, so the results were not statistically significant, but the trend was that the students who had the opportunity to use the VR system (especially those with access to the annotated environment) performed better than the control group, supporting the researcher hypothesis that VR could indeed be used as a supplemental tool to enhance learning [Bowman, Wineman, and Hodges].

Later, in the usability study, members of the class (in eight teams of three) were allowed to design their own habitat. Each team began with the actual layout of the habitat. After a brief acclimation period, one team member was selected to be the "design implementer" and work in the virtual environment. The others would give suggestions as they watched the computer monitor. Designs took about 30 to 60 minutes to complete, and still screen images were saved of the new layout so the students could present their design to the class. The VR developers recorded the errors and comments relating to the system made by the users, as well as surveyed the users before and after the experiment. The overall result was that in a short amount of time, students (the users) were able to alter an architectural space, putting their own aesthetic mark on the environment [Bowman, Wineman, Hodges, and Allison].

Conclusion

Overall, the developers feel good about the future of educational uses of VR. The exhibit can be used by students in preparation for a field trip to the actual zoo, or in cases where it is difficult for children to visit the zoo, in replacement of a field trip. The system was not designed to be travel worthy. Durability would be a key concern for creating a road show–style display that could travel to schools, libraries, and other zoos. With proper design and practice, a touring exhibit could be setup in less than an hour.

Another possible enhancement would be to include imitating the behavior of other types of animals. The developers were approached by the Grizzly Discovery Center located near Yellowstone National Park for a bear exhibit, as well as the Ontario Science Center and a German center interested in exhibiting VR experiences.

Project concept and organization was done by Larry Hodges and Jean Wineman. Don Allison designed and implemented the overall system, and created the gorilla models. Habitat and building models were by Brian Wills, and the annotation system was done by Doug Bowman. From the zoo were Kyle Burks, Kristen Lukas, and Lori Perkins, who all provided the gorilla information, and zoo director and Georgia Tech professors Terry Maple and Craig Piper, who provided access to the zoo facilities, the building used for the trial run, and some funding. Additional funding was provided by EduTech, and the Atlanta SGI field office provided the Onyx RE2 and Indy workstation. The overall effort took about 9 months from the initial work until the first public demonstration. The *Virtual Habitat* application was developed by Doug Bowman also with input and advice from Hodges and Wineman. Wineman and Maple's class on the "Psychology of Environmental Design" was used as the test case, providing subjects and evaluations of the system.

Both applications described here are designed for educational uses, outside the study of computer science or virtual reality. Thus for both to be integrated into curricula, or as part of an informal science exhibit (e.g., at a zoo or science museum/center), they need to be made robust and inexpensive enough to be practical. This is particularly true for the "society" application, which faces the onslaught of young children. To deploy the system at a venue, the

developers had to make the system more robust, autonomous, and easy to use. One specific change made to make this particular experience more feasible for a public venue was to remove the necessity of having a domain expert on hand to teach each user.

In April of 1998, a nonprototype version of the gorilla society application was deployed at the new Action Resource Center (ARC) at Zoo Atlanta for use by their guests. However, one of the greater concerns facing the Zoo was the logistics of how to demonstrate the application. The zoo has over 800,000 visitors a year, which is far too many to allow each one (or even a relatively small percentage) to experience the VR application first hand. The Young Scientists program was one group that is allowed to experience the application, as are zoo members and VIP visitors.

The installed system utilized a Dell XPS system with a 333MHz Pentium II with 256 megabytes of RAM, and an AGP Diamond FireGL 1000 Pro graphics acceleration card. A Polhemus InsideTrak electromagnetic tracker system with a single receiver was used for tracking the head.

A feature of the setup that called for improvement from the perspective of being usable by zoo visitors was to allow more group participation. This could be addressed both by having more systems and by allowing users of all the systems to interact within the virtual world, but also by allowing the people not wearing the HMD to participate in the activity. If 30 students visit the VR facility, each getting 5 minutes to wear the HMD, they end up waiting around for 2½ hours just for this short VR experience.

The developers believe that the virtual reality gorilla experience helped the zoo expand its message. Zoologist Lori Perkins explains it this way:

By combining Georgia Tech's advanced technology with Zoo Atlanta's advanced understanding of gorilla behavior, we're able to make the educational experience more powerful and dynamic. We want students to learn from both the live gorillas in the naturalistic zoo habitat and from the virtual experience, in order to give students a way to get a richer understanding of gorilla behavior than they would get from either experience alone.

CASE STUDY 6.5: TRAIN TO TRAVEL

The Dayton Bus Ride

For cognitively handicapped individuals, tasks such as using a city bus to get to a desired destination can be difficult. Many are capable of performing this task once they have received specific training for a route they need to take. The 1990 Americans with Disabilities Act (ADA) requires that public transportation authorities address the issue of how to make their system available to all. There are some training programs available to help handicapped individuals, but learning to ride can be time consuming for many individuals, making these programs (when they do exist) inefficient. Often the transit authorities resort to using door-to-door service, which is costly, and less than efficient.

In Dayton Ohio, the Dayton Public School system had a job training program for the cognitively handicapped, and part of the skills taught included bus travel training. However, the time necessary to train each individual meant that they could not meet the needs of everyone. To reduce the need for the door-to-door service, the Miami Valley Rapid Transit Authority sought a better solution for training cognitively disabled riders.

To help reach this goal, the transit authority funded a research group at the University of Dayton Research Institute to develop a multimedia and VR system to train handicapped individuals to use the bus system. Researchers Lyn Mowafy and Jay Pollack led the research project titled *Train to Travel*. While this project focused on travel training, the developers were also interested in using VR for job training in general for the cognitively handicapped.

Application Description

Train to Travel was comprised of two separate stages. The first stage used a desktop multimedia presentation to verify that each student had the skills necessary for the basic tasks required for riding the bus. Once the basic concepts had been mastered, the student would graduate to riding the virtual bus, and later a real bus.

The two stages were designed to work together using similar audio and visual scenes. Cognitively handicapped individuals learn better when repetition and familiarity are used as tools for training. Lyn Mowafy worked with Michael Stang (a school teacher with the Dayton Public



FIGURE 6-27

In Dayton, Ohio, a virtual reality simulation for bus travel was used to train students with disabilities in using public transportation.

School system) to ensure that there was a smooth and familiar transition from one learning stage to the next.

The target audience for this research project consisted of juniors and seniors studying in the Dunbar High School job-training program. Many of them had attained jobs at the University of Dayton. The trip between the high school and the university required a transfer between buses in downtown Dayton. In teaching these students to make this trip, the designers had three goals for the *Train to Travel* project. First they desired to incorporate training methods that were already proven in the public schools. Secondly, the application needed to be able to be modified for individuals of different capabilities and needs. Finally, the implementation needed to be implemented on readily available computers to provide classroom-based alternative travel that was effective, safer, and more efficient than individual mentoring of students in the real world. [Mowafy and Pollack] The first (pre-travel training) stage taught and tested for six basic skills necessary to ride the bus using a multimedia system:

- how to properly dress for riding the bus,
- how to identify and count money, and produce the appropriate fare,

- how to tell time so they would know when to catch the bus,
- how to select a seat on the bus,
- how to identify their destination, and
- how to find the departure points.

There were also two special modules for learning how to handle emergency situations, and preparation for riding the virtual bus. Each of the module's lessons was taught by a digitized tutor. The tutor was an actor hired to present the material. The tutor appeared with the tutorial information as a talking head in a window on the screen.

Once all the multimedia lessons were mastered, the student was ready to take the virtual bus ride. One of the concerns of the student's teachers was that the HMD might induce epileptic seizures. However, after testing the system with students who were subject to such seizures, there was no indication that the system was triggering seizures even when the students used the full system for the full virtual bus ride. In fact, the students were quite excited to be using such high-tech equipment and found the experience to be enjoyable.

The actual ride from the high school to the university took approximately 45 minutes. However, the virtual bus ride distills this to an 11-minute journey, covering only the crucial portions. In a full-length ride there were long stretches of time in which not much happened, and was considered not relevant to the training. To allow the student to faithfully experience the complete ride would have required considerable effort to create the virtual world, and it was felt that this effort would "greatly exceed the training benefit." The "long boring stretches would also not engage the student in meaningful interactive learning."

The portion of the ride that was included was from the point of seeing the important landmarks that indicate the bus is about to their transfer point, and on the second leg, the landmarks they should see just before their final destination.

To ride the virtual bus, the student dons the HMD, and is given a computer mouse. In riding the virtual bus, the familiar tutor from the multimedia experience would guide the student through the steps of boarding, paying, and finding a seat on the bus. During the course of the ride, the tutor



FIGURE 6-28

During their virtual bus ride, students passed familiar landmarks. Training students to identify these landmarks was a key goal of the program.

reminds the student to look for specific landmarks. On the way to the transfer point, they watch for the Spaghetti Warehouse after seeing the Sinclair Community College. They exit the bus at the Spaghetti Warehouse, and transfer to another bus and complete their journey to the University of Dayton's Miriam Hall after seeing the Frisch's Big Boy restaurant on the corner.

A mouse was used by the student to control when they would get on and off of the bus. When the student chose to enter the bus they would be "automatically transported into the bus and set in a seat." There was no need to ring the bell on the bus to request a stop since the way-points were at designated stops on the route, so all the rider needed to indicate was when to get off. The developers planned to allow the student to control their travel from one bus, down the street, across the intersection, to the next stop, and also to allow them to choose and move to their own seat. However, this method was not implemented, and instead a ride-along whisked the student from the bus to the next stop.

The virtual world was created by obtaining maps of the Dayton bus system, and using digitized photographs to create realistic-looking landmarks along the course of the bus in the parts of the trip encountered during the 11-minute

**FIGURE 6-29**

Students used an HMD and handheld controller to interact with the application.

edited version of the trip. The only sound in the virtual world was that of the tutor's voice.

Mike Stang, the job-training coordinator for Dayton's public school system, worked with Jay Pollack to come up with the basic ideas that evolved into the *Train to Travel* program. The overall design of the training system was done by Lyn Mowafy and Jay Pollack. Mowafy was the technical lead for the project and provided the principle design of the system. The overall project involved a team effort, including the help of programmers from Division Inc. VR programming was done by Andy Hamilton, Steve Hammond and Tom Congdon. The multimedia tutorials were implemented by Andy Spence, Mike Bibbey and Trish Russo.

VR System

A Macintosh computer served as the host for the multimedia training sessions. The model of the virtual world was created on a Silicon Graphics Onyx™ using Designer's Workbench™ and EZT™ by Coryphaeus. Upon completion of the model the project was transferred to a Division Inc. ProVision 100™ for the virtual bus ride. DVise™ by Division was used to animate the bus ride, orchestrate the audio components of the companion/tutor, and allow student interaction using a 3D mouse input device. The ProVision 100 VPX™ head-mounted display that was packaged with the system was capable of 320 × 240 color pixel resolution with stereoscopic display. Audio was rendered by storing

sound samples in an AKAI™ MIDI unit connected to the Division system. MIDI commands from the program would trigger the playback of specific samples.

The interactive multimedia tutorials were created on a Macintosh Quadra 950™ equipped with a SuperMac DigitalFilm™ board for full-screen video production. A standard NEC 4FGe™ color monitor, keyboard and mouse complete the system. Software packages used in the production of the tutorials included, MacroMedia Director™, Adobe Premier™, Adobe Photoshop™, Specular Infini-D™, MacroMedia SoundEdit™ Deneba Canvas™ and MicroSoft Word™.

Conclusion

A major benefit of the *Train to Travel* system was to demonstrate the real possibility in how the time and resources necessary to teach a cognitively handicapped person to be more self-sufficient by riding the bus to and from work could be significantly reduced. However, only anecdotal evidence is available to support this belief. Tests of the prototype system indicated that the number of real trips necessary to learn the route was reduced. Despite not having the opportunity to do a full study on the system, using a large population of potential beneficiaries, presentations of the system to travel trainers, teachers and transit officials met with resounding optimism that the *Train to Travel* system could revolutionize how this "very costly and intractable problem" is addressed in the future.

**FIGURE 6-30**

Users were provided with a first-person perspective inside the virtual bus.

Another benefit of using VR as the student's first encounter with riding the bus is that the student seems to be more independent from the outset. According to Mowafy and Pollack the presence of the instructor can be detrimental in real-world training situations because it "either modifies the experience or prevents it from occurring at all" [Mowafy and Pollack]. The virtual reality method also provides a means for the parents or guardians of the students to witness their performance in the virtual bus ride, prior to attempts in the real world. Seeing this helps to alleviate some of their fears.

According to Carla Lakatos of the RTA, reducing the number of people who must rely on the door-to-door services

allows them to better serve the people who do need these services. According to Stang giving cognitively challenged people the skills to independently travel also helps to raise their feeling of accomplishment and self-worth.

There were several features that were planned to help enhance the realism of the virtual bus ride. One such feature was to add interactive passengers to the bus. However, due to lack of funding, the project ended before these features could be added, and before a scientific study could be done to measure the effectiveness of the system.

Both of the lead developers feel that this system demonstrates the ability for using computers in general and VR in particular for training individuals with cognitive handicaps. Pollack points out the benefit of computer training over human training:

"We found that the students worked very hard when trained in this way. First the computer had infinite patience, never losing its temper after the tenth or 100th try. And very important was that the training gave each student a sense of pride since they 'got to use computers too, so cool.'"

Mowafy declares how the improvement of technology since the time they first attempted this project has made it a far more feasible use of technology:

"It was a great idea ahead of its time. Today, we could do this so much better. The major problem with projects like these is funding. Our society needs to value all of its human resources and provide them with the opportunities to be productive, responsible citizens."

CASE STUDY 6.6: FORTRESS OF BUHEN

Executive Summary

Virtual travel offers many new possibilities and advantages over physical travel. Advantages include elimination of transit time, plus cost savings once the virtual reality technology matures. However, the primary advantage of virtual travel is the ability to visit places that are otherwise inaccessible.

There are several reasons why a particular location might be inaccessible, including high degree of danger, high expense to the traveler, or restricted access for legal, religious, or other reasons. The most extreme reason is that a destination is inaccessible because it no longer exists or has become buried. Many of these reasons apply to sites of archaeological interest, making archaeological recreations

**FIGURE 6-31**

The Fortress of Buhen makes an ideal destination for virtual travel. The site no longer exists in the real world. However, a digital recreation of the fortress is accessible via virtual reality.

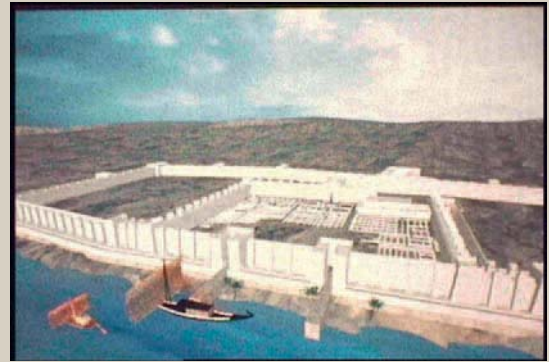
of ancient cities, temples, fortresses, etc., good candidates for utilizing the medium of VR for virtual travel.

The *Learning Sites Fortress of Buhen* project is one example of enabling travel to a place that is no longer accessible. Buhen is located on the Nile River near the ancient Nubian-Egyptian border. The Fortress was part of the Egyptian border control during the reign of Pharaoh Sesostri III. The ancient site is now submerged under Lake Nasser, created when the Aswan dam was built in 1964. A combination of factors makes the Fortress of Buhen an ideal candidate for a VR archaeological experience:

- The remains of the fortress are “extensive and well documented.”
- It is “no longer available for on-site investigations or first-hand visits.”
- “The site has a fascinating multicultural history dependent on economic and social cooperation.”

Introduction

The Fortress of Buhen was built on the west bank of the Nile and extended over 150 meters. It had a variety of interesting features, including “moats carved three meters deep into bedrock, revetments, drawbridges, bastions, buttresses, ramparts, battlements, embrasures, loopholes, and

**FIGURE 6-32**

The Fortress of Buhen was located on the Nile River and near the border between Nubia and Egypt.

catapult stations.” It was one among a chain of fortresses along the Nile, each built near enough to be within eye-shot of the next.

Some archaeologists believe that because it was located near the border between Nubia and Egypt, Buhen probably served as both a heart of commerce as well as defense. The fortress had many sections to serve different economic and social roles. There were separate buildings that probably served for administration, storage, and occupancy. An inner city within inner walls of fortification would have housed Egyptian settlements or those of the current occupying peoples. The area between the inner city and the outside fortification would have contained mercenary forces.

There is a paradox with respect to realistic portrayal of historical archaeological recreations. VR models can sometimes be criticized for being too “cartoon-like.” That is, being overly colorful and bright, and lacking the details of the actual physical structures. On the other hand, the use of texture maps can make a scene look somewhat photorealistic, and therefore can be criticized for being *too* realistic. A high degree of photorealism can be misleading to the virtual visitors, leading them to believe what they are experiencing is an exact replica of what once existed, whereas the model is more likely to be an interpretation synthesized from various bits and pieces of evidence, as well as some conjecture on the part of the experience developer.

When treated with proper consideration, a virtual reality representation of the site can actually be used to allay the misconception of which details are fact and which are speculative. A variety of data and representational styles can be merged into a system that allows the visitor to view the site from a variety of perspectives and interpretations. For example, a quick surface rendering of a reconstruction might have distinct visual qualities for data procured from different sources. Perhaps data gathered from photographic evidence might have more subdued colors, whereas the conjectured parts of the structure might be represented with bright “cartoony” colors. If multiple theories exist for how a site may have looked during its time of use, there could be an interface to shift between these views and a photographically based view of the actual remains.

Annotations can be used to provide the user with direct access to the source material, field notes, or reasoning behind particular interpretations. These annotations might contain images, sound, video, and text to help describe objects and provide background material on a particular aspect of the site.

A VR experience that provides such a multiple perspective view of the site, giving the visitor the opportunity to view the site from the eye of multiple archaeologists might lead the participant to a better understanding of the archaeological process.

The *Virtual Fortress of Buhen* re-creation used images texture mapped onto a simplified polygonal model to represent the structures of the site. Texture map images were used to mimic the visual surface textures of the mud brick coursing and stone brick walls, plus decorations on some of the interior walls. These images were constructed from drawings and photographs as well as descriptions from the reports of the excavation and discussions with the curators at the Museum of Fine Arts in Boston.

Rendered as a life-size model, participants are able to gain a sense of the enormity of the various spaces, as well as “the details of the decorations and building techniques” exhibited in the site.

Application Description

A visit to the *Virtual Fortress of Buhen* uses a tour guide metaphor to give visitors a foundation on which they can



FIGURE 6-33

A computer-generated agent acts as a tour guide aboard a pharaoh's ship of state.

begin their exploration. A computer-generated agent modeled as an ancient Egyptian scribe acts as a tour guide to show the participant around the virtual world. The tour guide follows a preset path, but the guest can look in all directions at any time, and can choose to follow the guide or venture on their own down other paths. No other interaction was possible with the tour guide.

The tour begins onboard a ship docked near the entrance to the fortress. The ship is an Egyptian pharaoh's ship of state of the period. Visitors can move about the Pharaoh's ship and accompanying cargo ships to experience them as they may have looked during the active days of Buhen. After exploring the ships, they are led onto a dock. At this point, travel control is taken from them and they are then “flown” around the outer wall, giving an overview of the site. After the ride-along flight, they are then led through the main gate of the fortress to tour the inner rooms of the city.

Representation of the Virtual World

In addition to providing the ability to seemingly experience an ancient site first hand, there are other benefits that arise from the creation of a virtual, digital copy of an archaeological site. One benefit is the ability to hold different interpretations of the site within a common context. Information about materials collected by, and owned by, different entities can be merged together into a single form. Storage in digital form also allows for the inclusion of annotations tied directly to the data, so whenever the data is accessed, a list of notes can be summoned automatically.

Digital information is also generally more transportable, and some formats allow it to be moved easily from one type of machine to another. Digital information does not gradually decay as the original source material does. The digital information can also be archived with the expectation of reasonable (though not infallible) durability.

Two other benefits arise from the ability to virtually travel. Sites or artifacts that cannot be visited or touched due to religious or other restricted access reasons can often be made accessible to the public in digital form. Also, the ability to telecommute becomes a possibility for some archaeologists. A lead archaeologist or expert on some particular topic can be consulted on various issues by archaeologists physically at a dig site.

There are some pitfalls with the digital world as well. First, there is a concern of whether the current data will be accessible on future computing platforms. Computers become obsolete fairly quickly, so data that is limited to specific platforms face the possibility of becoming unusable. The use of standard formats helps to alleviate this problem, but sometimes standards don't provide for storing sufficient detail. Also, information is often scanned and stored at a level of detail based on the technology of the day. Digital representations should be created at the highest resolution possible, with reduced resolution versions for rendering or interrogation by current computers, with the expectation that future computers will be able to handle more information. However, the limits of the scanning devices may require artifacts and the like to be rescanned at some point in the future anyway.

The other major pitfall is the legal issue of ownership. In the simpler case, archaeologists will probably not want to share any new data they uncover until after they've published a report based on it. After this, they will often be willing to allow their data to be incorporated into a global database about a particular site. On the trickier end of things, when there is a possibility of money being made from the data, people may be less inclined to share their information. This snag is becoming less of a problem as the "virtual archaeological" companies such as Learning Sites establish working relationships with the new generation of archaeologists.



FIGURE 6-34

A virtual tour guide points out an information kiosk in the virtual world of Buhen. The panel on the left shows a map of the site, the center panel displays a photograph of the actual site, and the right panel provides a text description of interesting features that are nearby.

To provide context on the *Fortress of Buhen* data, virtual kiosks are available in various locations in the virtual world. For participants interested in learning more about the site, the kiosks provide access to additional information from the digital databases. The concept is to provide the capability for visitors to explore the site as it was at any time throughout history (including points in time during its excavation) and allow them to access information on this and other sites from databases located around the Internet. Visitors could use the kiosks to make inquiries about the site at varying levels of detail, depending on how casual or engaged they are about the subject.

Interaction with the Virtual World

The kiosks in the *Fortress of Buhen* world had three panels of information. The left panel provided a map of the virtual world, with a "You are here" wayfinding aid. This panel was also intended to provide a view into a GIS database of the full site and surrounding region including geographic and topological information. The center panel shows a photograph of the actual excavation site from that spot. The right panel provides a text description of topics related to the nearby interesting features. The plan was to make it a full hypertext database of information related to this site, as well as other relevant sites, that the participant could jump to via a portal in the kiosk. The application authors

had also intended for the kiosk to make available a visual database of the artifacts and archaeological remains relevant to that area.

VR DISCOVERY

The application developers found it worthwhile to provide “information kiosks” in the virtual world from which participants could access further information about what they were experiencing from the World Wide Web.

Further Educational Uses

In addition to the specific archaeological uses of this technology, the developers of the *Fortress of Buhen* (specifically Learning Sites and ERG Engineering) see other educational uses of these applications of the technology.

The ability to appreciate other cultures is enhanced when students can see how the ancestors of various modern cultures lived. In the words of archeologist and architect Donald Sanders: “Archaeology reinforces the concept of a shared human heritage and provides modern people with perspectives on their own place and time in history.”

Additionally, an archaeological VR application provides students with insight into how a scientist uses fragments of information to build up a theory. Students can see how archaeology (and other sciences) is in fact an exploration into the unknown, with the scientist working to unravel the mysteries. Also, learning can branch off into concepts such as languages and mathematics by examining the languages of the ancient cultures, the mathematics used to construct buildings, or the astronomical information these cultures used to lay out structures in ways to take advantage of the movements of the solar system.

Another combined effort between Learning Sites and ERG was to make an archaeological application specifically designed as an educational device. The result was the *Vari House*, a Web-based experience on an ancient Greek farmhouse. The goal for the *Vari House* project was to enable a broader audience to experience the site in three dimensions. Since their constituent audience doesn't generally

have access to high-end computers and VR hardware, a VRML solution was implemented. Although VRML models can be viewed via immersive displays, this application was designed for nonimmersive, desktop use. Using the desktop browser, the application developers combined textual information alongside the 3D visual models, allowing students to read about the world as they explore it. In addition to traversing the model of the grounds, the user can click on objects in the scene to see a detailed model of a particular artifact and/or access a textual database.

Venue

The *Fortress of Buhen* project was begun in 1993 by Bill Riseman. His overriding archaeological goals included painstaking attention to scholarly standards and a high level of detail. He was also interested in enhancing the experience by utilizing 3D models, audio, text, still images, and video, as well as enabling multiple participants to interact and share within the world. He formed a company and assembled a team consisting of archaeologists, 3D modelers, and virtual reality experts to bring about his vision. In particular, he worked with archaeologist and architect Donald Sanders and virtual reality application designer Eben Gay.

Virtual reconstructions of the *Fortress of Buhen* were based “on primary evidence and documentation (excavation notebooks, field photographs and drawings, and early travelers' views)” [Sanders]. Standard computer-aided design and drafting packages were used to create the 3D forms used in the computer renderings. Following this, details were added by texture mapping images onto the polygonal shape of the structures. At each stage of reconstruction, the models were validated by seeking out the opinion of expert archaeologists. Once the models were created, the interaction and narrative features of the experience were added. These features include the information kiosks and the guided tour elements of the experience.

Much of the documentation used to create the models and informational databases used to construct the *Fortress of Buhen* was provided by the Egyptian department of the Boston Museum of Fine Arts. This information included:

- Handwritten field diaries
- Progress photos of the excavations
- Excavation reports (including stratigraphic data and building plans)

- Topographic survey plans
- Hieroglyphic translations

In addition to the museum-supplied documentation, Riseman developed a new procedure for rendering a more accurate representation. The procedure, called by Riseman *epigraphic rendering*, used a computer program to automatically trace objects in a photograph, making the look of the resultant image more correct and easier to use for translation. As the experience toured various venues in 1994, Riseman and his colleagues found resistance to accepting his new method. Having been trained and grown accustomed to more traditional methods, archaeologists were reluctant to utilize different methods than they were used to. Tragically, however, before the project was completed, Riseman was accidentally killed in Brazil, and some of the original ideas were not completed in the *Fortress of Buhen* experience. Development on the *Fortress of Buhen* project abruptly ended with Riseman's death due to legal concerns with intellectual property ownership.

In late 1995 Donald Sanders created a new company (Learning Sites Inc.) and constructed a team to continue the work he started with Riseman. Collaborators in the new team included Tim Kendall, an archaeologist from the Museum of Fine Arts; Veronica Panttelidis, a teacher and head of the VR lab at East Carolina University; VR consultant Eben Gay from ERG Engineering Inc.; and 3D modeler Rick Morse. ERG Engineering had considerable experience in developing virtual reality and desktop virtual world products designed for education and for museum venues.

VR System

The application was originally developed on a Kubota-Denali computer. At a later time, the system was moved to an Evans and Sutherland Freedom 3000™ image accelerator with a Sun Sparcstation acting as the host. The visual display was provided by a Leap Cyberface 3™, which is a head-based display similar to a BOOM™. Software was written using the *Sense8 WorldToolkit™* to develop the interactive virtual world.

The *Fortress of Buhen* was demonstrated at many venues throughout 1994, including the SIGGRAPH computer graphics conference held in Orlando, Florida. However, the original files were lost, and it is no longer possible to

experience it first hand. The developers (Learning Sites and ERG) have recreated VR representations of other sites, each with their own special capabilities and characteristics, but bearing similarity to the Buhen experience. Among the other sites built are a tomb at Nemrud Dagi, Turkey, and Temples B300 and B700 at Gebel Barkal.

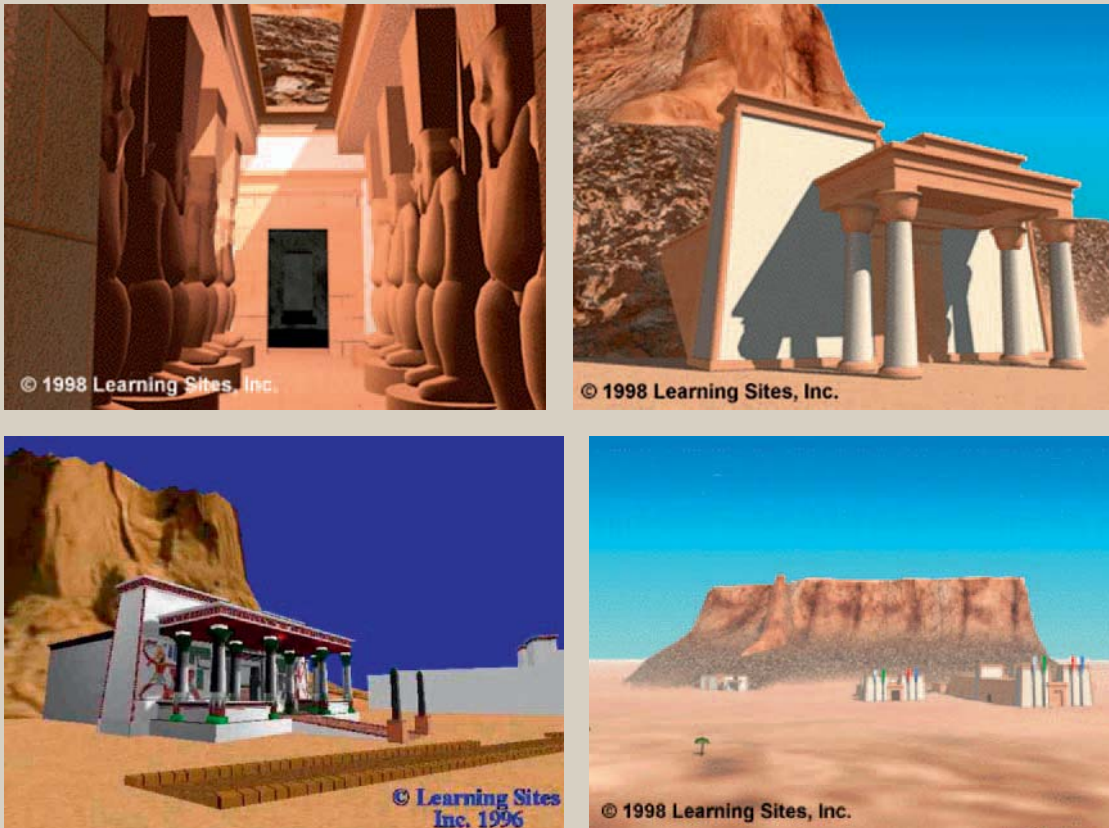
The tomb at Nemrud Dagi is enhanced through the medium of VR because the setting of the tomb is a breathtaking view that can best be felt by experiencing it firsthand, from the first-person point of view. The temple at Gebel Barkal is interesting because it is a currently active archaeological dig that is being used in an experiment to combine the Internet and VR to enhance the archaeological research.

Application Implementation

A JAVA-based search engine allows researchers to find information within all the documents. Active hyperlinks provide the means to jump to the excavation context in which a particular artifact was found or to which a written note refers. A 3D virtual world can be created specifically for the needs of the user. A user can select artifacts and references of interest, and a model of the world is constructed specifically for examining those specific interests.

This project could be used both as a research tool for archaeologists and for educators. Archaeologists can study the site as if they were there, jumping between excavation notes and the representation of the site. The tool can also provide the means for teaching a college-level course on archaeology. Students could freely explore the site and the accompanying information, and later be queried for possible hypotheses on how the site may have looked or how particular artifacts may have been used by the residents of the site.

Other worlds were created for educational purposes using VRML models available via the World Wide Web as onscreen walkthroughs. The Vari house is one example of the use of the Internet to provide an educational web site based on archaeology. A curriculum was developed by which students could individually explore the VRML model of the excavation, with an accompanying text description of the model followed by questions that require exploration of the site to find the correct answers. Artifacts in the site can be triggered to produce a separate rendering, with additional textual information about the particular item.

**FIGURE 6-35**

These four images show different aspects of a subsequent project that depicted different temples at Gebel Barkal.

ERG Engineering also demonstrated other virtual reality applications with interface features that could be applied to museum and other educational venues. In 1995, they demonstrated an augmented reality (AR)⁸ application at a trade show that allowed viewers to see additional information about the trade show exposition as they explored the premises. As they looked toward different vendors, the name of the

vendor was displayed above each booth, plus a short description of what each vendor sold. In this instance, the AR technology was demonstrated primarily as a navigational tool to help people locate their destination. Such technology could also be applied to applications where someone could look at an exhibit, site, or object on display and individually request specific information on what they purview depending on their level of knowledge or interest in the particular topic.

⁸Augmented reality (AR) is the paradigm of display in which the user can see the actual real world in registration with additional overlaid computer supplied information.

Conclusion

The *Fortress of Buhen* archaeological experience demonstrated how many features of virtual reality can be beneficial

to the science of archaeology, as well as be an educational tool using archaeology as the content around which the students learn. It also demonstrates quite clearly how VR can provide a means of travel to a locale that is no longer possible to visit.

Learning Sites Inc. later developed tools to explore the possibilities of using this technology to do “teledigging,” that is, allowing researchers on site at a dig to enter information, providing access to experts in the field located

elsewhere. Thus, the experts can give comments and recommendations to the onsite researchers.

Another possibility that became practical thanks to VR and new visualization techniques was the ability to use new sensor technology that could gather information about artifacts or the earth itself using non-destructive sensing. Some of this data required techniques such as volume rendering to represent the contents.

CASE STUDY 6.7: VIRTUAL MOBILITY TRAINER

Dr. Dean Inman, senior scientist at the Oregon Research Institute (ORI), is driven by a desire to help children with disabilities. Though he was intrigued by the notion of virtual reality, he initially doubted whether VR could really be of practical benefit to his disabled patients. Then, fortuitously, in 1993 his wife, Dr. Lynne Anderson-Inman, the director of the Center for Electronic Studying at the University of Oregon, heard Brenda Laurel (see Understanding Virtual Reality) speak at a conference on interactive technology. She saw that VR had potential, and also learned that suitable technology was becoming available at reasonable price points. She felt inspired and conveyed these ideas to her husband.



FIGURE 6-36

A child learns how to control her wheelchair with the aid of a virtual reality system.

Image courtesy the Applied Computer Simulation Labs

Inman had been studying the problems of teaching children to operate motorized wheelchairs for several years. There were several difficulties associated with this effort. One major problem was that it was difficult for many students to get access to motorized chairs. The chairs are very expensive, and insurance companies won't purchase them unless they are assured that the child is likely to have success with the chair, as indicated by their ability to manipulate them. Thus it is a paradoxical situation that the students need a chair to learn but can't get a chair until they have demonstrated that they can skillfully operate one. Failure of a child to learn to explore the world around them is known to be detrimental to their ability to analyze spatial problems, among other skills.

Inman began a program to teach children with cerebral palsy to operate motorized wheelchairs in 1982. Initially, this “driver's ed for wheelchairs” was done completely by putting the child in a wheelchair and closely attending to and giving the child instructions on what to do. The program later attempted to use computers (Apple IIe and later an early Macintosh) to display a simulated forward view as a means of raising interest and motivation in the children. However, this did not turn out to be a viable solution.

A significant problem is that many of the children are not motivated to learn. Without their own mobility they are attended to and waited on. Thus children may not perceive that an increase in ability for independent travel is beneficial to them. The lack of motivation on the part of the

potential beneficiaries of a program that offers the chance for increasing one's ability to independently move through the world can at first be puzzling. This can at least partially be attributed to the fact that they become accustomed to receiving aid for many daily tasks. Inman refers to this as "learned helplessness."

After becoming excited by the potential of VR, Inman wrote and won a \$600,000 3-year grant from the U.S. Department of Education to build a system to train disabled children to drive motorized wheelchairs in a virtual world that emulated driving motorized wheelchairs in the real world. The grant began in July 1993, at which time Inman began to build the *Virtual Mobility Trainer (VMT)* to help train and motivate children in the use of wheelchairs. He subsequently received other grants to continue this research and expand in new directions.

Application Description

The efforts funded by this grant resulted in a virtual reality system for use as a training simulator. In the *Virtual Mobility Trainer* system, children wear a head-mounted display to view the virtual world, and sit in a conventional wheelchair mounted on rollers with motion sensors. The child operates the wheelchair with its normal controls and the platform calculates movement through the virtual world by measuring the movement of the wheels.

In his previous efforts to incorporate a computer into the training, a particular wheelchair was selected and physically integrated into the system. The choice to use a platform capable of working with any wheelchair was made for the *VMT* because it allows students who already have their own wheelchair to use it as the interface to the VR system. Another benefit of placing a fully functioning wheelchair on a platform with rollers is that it allows the actual motors on the wheelchair to run. When the actual motor and wheels run in response to the input from the user, the user is provided with the vibratory and aural sensations consistent with what they would normally experience when operating the wheelchair in the real world. This effect is heightened by using rollers with bearings that are slightly off center which give the sensation of riding over moderately rough terrain such as a sidewalk or carpet [Inman et al.].

VR DISCOVERY

Inman's team found a solution to simulating inertia realistically. They simply added ballast to the rollers in the platform. By raising the weight of the rollers from just a couple of pounds to 25 pounds, the wheelchairs placed on the platforms would have to overcome the inertia of the rollers to get moving at higher speeds. This more closely simulates the real world.

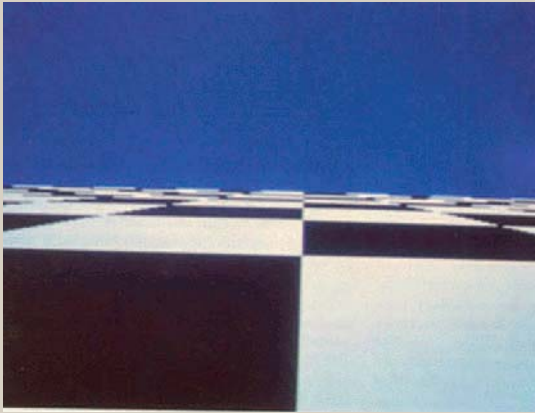
Three initial virtual worlds were constructed in which students could practice specific tasks:

- An empty VR training world
- A large abstract space with interesting objects
- A model of a real-world crosswalk

The first virtual world is a simple, wide-open space with an infinite checkered floor. There are no obstacles in the environment. The main goals for this world are to learn the basic "go," "no go," "keep going," and "stop" operations. The students can also practice turns, stops, and reversing. The system can emulate the visual effects of traveling very fast, which is exhilarating for many of the children. This world is intended to allow the student to learn to manipulate the chair, and experience the sense of freedom brought about from driving a chair by oneself.

The second world is a model of a space a quarter of a mile long and a quarter of a mile wide containing ice, mud, and obstacles to avoid. The primary learning goal for this world was obstacle avoidance. Throughout the grassy world there are 3D objects and patches of mud and ice that cause the driver to get stuck or spin out of control. The 3D objects are represented as polygonal geometric shapes. If the child hits an object, a crash sound is activated, and the child cannot move the chair until they back up and turn away from the object.

In order to keep the exercise entertaining and engaging, a scenario including interactive animation, 3D sound, and simulated "hazardous" surfaces such as ice and mud was created. Some of the objects in the world are targets to drive over, with success rewarded by the display of a

**FIGURE 6-37**

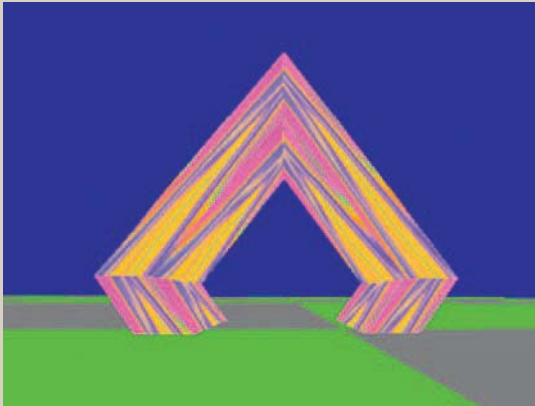
A simple checkerboard virtual world.

Image courtesy the Applied Computer Simulation Labs

**FIGURE 6-39**

Learning to cross a street in VR is safer and much less stressful than in the real world.

Image courtesy the Applied Computer Simulation Labs

**FIGURE 6-38**

Fun objects in the virtual world provide interesting things to do.

Image courtesy the Applied Computer Simulation Labs

waterfall, birdcalls, a performance by a popular entertainer, or some other fun event. If the child travels far enough, they arrive at the edge of the world and can fly by going beyond the edge. The goal of this world was to demonstrate that exploring a new space could be enjoyable. The kids found it particularly enjoyable to go through the mud and ice.

The final world was a model of an actual intersection in Eugene, Oregon. The world contained cars, crosswalks, and lights. The goal was to teach the child the real-world task of crossing the street safely. Training this task in VR is much less stressful than doing it when the child first encounters a real intersection. Part of the research study was to measure the effects of the VR training in a real-world activity.

In this realistic world, it is very important that the children not be taught to enjoy hazardous situations. The application developers expended a great deal of energy into making sure that if the child were struck by a virtual vehicle, the result was *not* fun.

Sound was integrated in the virtual world as musical themes and abstract tones. The sounds were associated with visual objects, and could be heard only when in proximity to the objects. The children enjoyed the wide variety of sounds.

It is important to note that none of the sounds were directly correlated with actions in the virtual world, other than that they were associated with locations. That is, there were different sounds emanating from different objects, and they were only heard when the student was near that object. However, the sound was not necessarily directly related to the object it was associated with.

The first year of the grant was spent developing the system. The second and third years were focused on evaluating the kids that use it. The results of their study indicate that the VR wheelchair training system is effective in allowing children to gain and increase their skill in operating motorized wheelchairs. The more time children spent exploring in the virtual world, the more their skills improved.

For the first half of the grant period, subjects were required to visit the ACSL lab in Eugene, Oregon. However, a year and a half into the project the developers made a satellite facility available in a public school in Portland to make the study available to a greater number of children in the target population. Though some children were only able to participate a couple of times, others used the system dozens of times. The length of each experience varied from 15 minutes to 45 minutes per session, with an average usage of 30 minutes.

The main difficulty in training these children is maintaining their motivation, particularly in light of the fact that it can take considerable time and effort to get them to the facility. This effort caused many people not to participate, and others to drop out early, having not found the virtual worlds to be interesting enough to make the trip. The diverse group of participants demonstrates that the *Virtual Mobility Trainer* system can be beneficial to a wide population. This study has also pointed out the limitations of VR for this task.

Although the system was designed for and is capable of displaying stereoscopic imagery in an HMD, the developers found that many children prefer to view the experience monoscopically on a large monitor.

The VR System

In 1993, the system Inman's team built consisted of an Intel-486 based personal computer with two SPEA Fireboard™ graphics accelerators. The dual graphics accelerators allow images to be rendered for each eye to produce a stereoscopic view. An Eyegen3 HMD from Virtual Research was used to display the 3D world, with the head tracked by a Logitech ultrasonic tracking system. Three-dimensional audio was provided via a Beachtron 3D sound system. The headphones in the HMD do not occlude real-world sounds, so the student can converse with the instructor, and can

hear the sounds of the motors. The primary software was the *Sense8 World Tool Kit (WTK)*. In all, the system cost was about \$30,000.

The wheelchairs were mounted on a modified wheelchair athletic training platform. Modifications include ballast in the rollers and the addition of optical encoders to measure the movements of each roller. In the pre-VR setup, the students sat in a Fortress™ wheelchair coupled to the computers via an interface built specifically for this application. In the *VMT* setup, any wheelchair could be placed on the platform, but it was not directly connected to the computer. Therefore, the motors and brakes could only be controlled by the user and did not respond to the virtual world.

In the *Virtual Mobility Trainer* system, the frame rate of the simple worlds was about 25 frames per second. The more complex worlds, made up of about 750 polygons, were rendered at a rate of about 10–12 frames per second. A system today would be capable of rendering a scene with many more polygons and with more detail, at a far lower price.

In the street crossing training, a physical button was present for the children to press to activate the crosswalk indicator. The button provided an actual haptic response and resistance to being pressed.

One of the difficulties of the project was integrating the various devices from different vendors. The developers had to learn to put things together themselves because there was not a ready-to-run wheelchair training product on the market. Thus they became designers and system integrators to build the hardware and software system, and also to create the virtual worlds themselves. One thing they discovered as they implemented the project was that many different skills are required for building a VR system. These skills include knowledge about interactive 3D computer graphics, and writing interface software for specialized hardware input and output devices.

Even with their early success there were some areas they knew needed to be improved. In particular, lag and frame rate were suboptimal, and the joystick was not as responsive as desired. High lag in the tracking system resulted in delays between the time the child turns their head and when the appropriate view is displayed. Also, the HMD was cumbersome, weighed more than they desired, and had

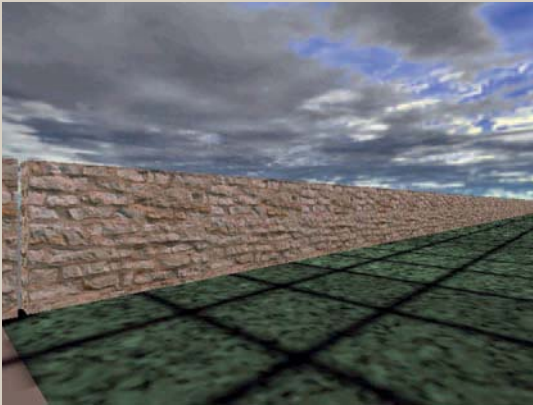


FIGURE 6-40

Advancement in computer hardware allowed worlds to be rendered more richly.

Image courtesy the Applied Computer Simulation Labs

less resolution than they felt was necessary. In fact, some children chose to explore the world without the physically immersive, but also encumbering, head-mounted display. They chose to view the world on a nearby monitor instead.

Part of the subsequently funded follow-up project was to upgrade to a system that would take advantage of the trend of improved computer hardware to make a system that would be more cost effective, and be able to display more interesting worlds. The later system used a 400Mhz Intel Pentium CPU system with no graphics acceleration hardware. Later, they used software developed in-house based on the Microsoft *DirectX™* rendering system.

The principle investigator on the *Virtual Mobility Trainer* project was Dean Inman. Ken Loge designed the graphics

and sounds for the initial system and designed the simulation environments. He served as the manager and chief designer for the project. John Leavens and Aaron Cram programmed the systems, and Cyrus Kanga worked on the 3D and 2D graphic design.

Conclusion

Inman originally considered VR, as many do, a solution looking for a problem. He now feels that he has discovered at least one problem for which it is well suited. VR enables the team to provide a feeling of freedom and to improve the sense of space and direction to physically disabled children, as well as to teach them to use a motorized wheelchair. This helps those children feel more independent and gain more self-esteem.

Although Inman considers VR a useful tool, he does not consider VR to be an end-all panacea for solving the problems of the disabled community. He sees potential and is interested in exploring further how it can be a useful tool for the disabled population. However, merely being VR is not enough to make an experience interesting. It is noteworthy that while the novelty of VR was a motivating factor for getting children to participate in the “drivers’ ed” program, VR alone is not enough to sustain the interest. The content of the VR experience must be designed to hold the interest of the participant.

On the other hand, the mass media was overwhelmingly interested in the project to the extent that interaction with the media consumed a great deal of Inman’s time. Inman views the fact that his work has received considerable press as an indication that the public is interested in the idea that technology can be used for “a good, positive, uplifting purpose” [Buckert-Donelson].